# A fixed-target approach to the sharp iid Berry–Esseen inequality*

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**Abstract**

We give a conditional reduction of the conjectured sharp iid Berry–Esseen inequality and prove a strict comparison theorem for an explicit family of laws close to Bernoulli laws. At a fixed index, the excess above $C_{\mathrm{E}} = (3+\sqrt{10})/(6\sqrt{2\pi})$ attains its maximum. First variation and integration against the zero-bias density give a conservation identity containing a nonnegative defect that vanishes exactly at the extremal Bernoulli law. Under the sharp bound at the preceding index and the stated finite-verification conditions, this identity yields quantitative localization bounds. For $n \geq 700$, these bounds place every hypothetical extremizer in a common probability and support region. Conditioning on the number of positive summands then gives strictly negative excess: one count controls small conditional noise, while five neighboring counts and one concave quartic control the remaining range. The comparisons include threshold atoms and all incoming tails. The global sharp inequality follows if the sharp bound is established for every index through 699. That finite base is not proved here.

## Contents

1 The problem, results, and proof strategy . . . . . 2
2 Fixed-target extremizers and contact . . . . . 4
3 The two-point skeleton and its moment defect . . . . . 7
4 Conservation and control of the defect . . . . . 10
5 Quantitative localization . . . . . 13
6 From small defect to a common support region . . . . . 14
7 The direct comparison with a binomial law . . . . . 19
8 Composition and the finite boundary . . . . . 27
9 Discussion . . . . . 28
A A uniform Fourier enclosure with explicit input hypotheses . . . . . 29
B Elementary Gaussian constants . . . . . 34
C The identity without contact . . . . . 36
D Details of the coarse localization . . . . . 37
E Certificates and independent reproduction . . . . . 50

*This paper was written with substantial assistance from AI systems; see the end of Section 9.

# 1 The problem, results, and proof strategy

Let $\mathcal{P}$ be the laws on $\mathbb{R}$ with mean zero, variance one and finite third absolute moment $\rho(P) = \mathbb{E}|X|^3$. For independent copies $X_1, \dots, X_n$ of $P$, put $S_n = \sum_i X_i$ and

$$\Delta_n(P) = \sup_{H \in \mathbb{R}} |\mathbb{P}_P(S_n \le H) - \Phi(H/\sqrt{n})|, \qquad C_n = \sup_{P \in \mathcal{P}} \frac{\sqrt{n}\,\Delta_n(P)}{\rho(P)}.$$

Here $\Phi$ and $\phi$ are the standard normal distribution function and density. The sharp-constant conjecture is

$$\sup_{n \ge 1} C_n = C_{\mathrm{E}}, \qquad C_{\mathrm{E}} = \frac{3 + \sqrt{10}}{6\sqrt{2\pi}} = 0.409732\ldots. \tag{1}$$

Throughout, $H$ is a raw sum threshold; terminating decimals in inequalities are exact rationals. Affine standardization gives the equivalent result for any mean and positive variance, with third absolute central moment divided by the cube of the standard deviation.

Berry and Esseen established the classical order $n^{-1/2}$ [3, 5, 6]. Esseen's lower bound [7] explains the proposed constant. If $p_* = (4 - \sqrt{10})/2$, $q_* = 1 - p_*$ and $X = (B - p_*)/\sqrt{p_* q_*}$ for $B \sim \mathrm{Bernoulli}(p_*)$, the fixed-law lattice limit gives

$$\lim_{n \to \infty} \frac{\sqrt{n}\Delta_n(P)}{\rho(P)} = \frac{3 + q_* - p_*}{6\sqrt{2\pi}(p_*^2 + q_*^2)} = C_{\mathrm{E}}. \tag{2}$$

Indeed $p_*^2 + q_*^2 = 10 - 3\sqrt{10}$ and $(3 + q_* - p_*)/(p_*^2 + q_*^2) = 3 + \sqrt{10}$. Every constant below $C_{\mathrm{E}}$ therefore fails for a sufficiently large sum of this one law. The upper conjecture is associated with Zolotarev [23, 24]; Schulz proves the strict finite-index bound for every nondegenerate Bernoulli law [13].

**Fix the target rather than an unknown ratio.**

At a fixed $n$, maximize

$$a(P, H) = \sqrt{n}\{\mathbb{P}_P(S_n \le H) - \Phi(H/\sqrt{n})\} - C_{\mathrm{E}}\rho(P). \tag{3}$$

The strategy is a rigidity argument followed by a comparison with a Bernoulli sum. The moment penalty gives an attaining law; first variation produces a majorant that touches the conditional distribution function on its whole closed support. Integrating this contact relation against the zero-bias density [8] gives an identity with nonnegative terms measuring the excess, displacement of the threshold, and the defect

$$D = (3 + \sqrt{10})\rho - \gamma - 3J, \qquad \gamma = \mathbb{E}X^3.$$

Here $J$ is the weighted sum of squared support-gap lengths, defined in (10). The defect vanishes exactly at the oriented Bernoulli law in (2). An upper bound on the remainder in the identity therefore forces a hypothetical nonnegative-excess maximizer close to that law.

For $n \ge 700$, direct inequalities give $3/8 < p < 93/200$, inward support displacement below $1/250$, and total conditional noise variance below $7/20$ in count coordinates. Two contact inequalities bound the outward ends of both conditional supports. A single-count estimate for noise at most $1/100$ and a five-count estimate above it then give strictly negative excess. The shared endpoint and zero noise are included. The comparison theorem itself, Theorem 7.1, assumes only its explicit local law domain; it does not assume extremality, contact, or a predecessor bound.

**The finite verification.**

The computational part establishes a coarse localization estimate. The analysis places an attained nonnegative-excess maximizer, subject to the sharp bound at the preceding index, in an explicitly bounded moment region. A certificate records rational bounds for a finite subdivision of that region. The checking program reconstructs the subdivision and verifies that each part either contradicts a necessary condition or satisfies the claimed bounds. The parts cover the whole region, including their boundaries. Together with the analytical estimates, this proves in particular

$$nD \le 32.20665378, \qquad |H| \le 1.83605680$$

for every such maximizer and every actual $n \ge 42$.

Finding and checking the certificate serve different purposes. The exploratory calculation chooses subdivisions and candidate bounds; verification reconstructs the prescribed inequalities and checks their enclosures. The retained calculation uses rational arithmetic, with proved rational bounds for roots, exponential terms, and Gaussian constants. It controls whole parameter intervals, rather than inferring an all-law inequality from sampled examples. The analytical reduction and the coverage argument are parts of the proof; the data file alone does not replace them.

Subdivision and finite enclosures are established methods of rigorous numerical analysis; related methods have been formalized in proof assistants [17]. Computational bounding also has precedents in Berry–Esseen estimates [14, Section 2]. Our certificate format is specific to this calculation. Its Python verification is not a formalization of the entire argument, and its generation and checking share mathematical kernels. The later support and count comparisons have analytical proofs with short rational checks. Appendix E identifies the data, programs, arithmetic conventions, and separate reproduction commands.

**The finite boundary.**

The exclusion of an attained extremizer at $n \ge 700$ requires the sharp predecessor bound. A least failure supplies that bound only if all earlier indices have been controlled. A base through 699, together with the complete entry conditions, implies the sharp conjecture; Section 8 proves this implication. A base only through 41 locates a first failure in $42, \dots, 699$ but does not rule it out. Those finite bounds remain open in this work.

**Context.**

Fourier and structural moment inequalities progressively improved the universal upper bound [11, 14, 16, 19, 20]. We use Shevtsova's structural inequality explicitly at entry. Recent manuscripts report smaller upper constants [10, 21], and He and Cheng report eventual sharpness beyond a uniform, very large threshold [9]. Work on independent arrays and extremizers addresses the related non-iid problem [1, 2]. Tao's registry and the available formalization projects distinguish numerical certificates, analytical inputs and formalized implications [4, 18, 22]. None of these recent reported bounds is a premise below. An earlier AI-assisted working draft is on Hexagon [12].

## 1.1 Reading the argument

The proof selects an extremizer, identifies the zero-defect law, bounds the remainder in a conservation identity, and compares the resulting near-Bernoulli law with a binomial sum. Reflection selects the positive closed-event orientation before localization; its signed defect is estimated in that orientation. The finite Fourier calculation is used only to obtain the coarse

bounds. The subsequent support and count arguments are analytical, with explicit rational comparisons.

The following table distinguishes the three scales used in the proof. Gap-specific parameters are introduced after the existence of a central gap has been established.

| Symbol | Meaning and scale |
|---|---|
| $H$, $H/\sqrt{n}$ | Threshold for the original sum; its normal-standardized value. |
| $B_0 = \mathbf{1}_{\{X>0\}}$, $p, q$ | Sign class and its probabilities for the original marginal. |
| $m$, $\ell = m/(pq)$ | $m = \mathbb{E}X_+ = \mathbb{E}X_-$; distance between conditional means. |
| $W$, $\theta = \mathbb{E}W^2$ | $W = X - \mathbb{E}(X \mid B_0)$; conditional noise in original units. |
| $U = W/\ell$, $s = \mathbb{E}U^2$, $\lambda = ns$ | Noise in count units; marginal and total noise variances. |
| $t = np + H\sqrt{pq+s} = k + u$ | Count threshold, integer anchor, and phase. |
| $a_0, a_1$, $d = a_0 + a_1$ | Inward support displacements in count units. |
| $J$, $D$ | Weighted support-gap sum and defect of the original law. |

The sign decomposition is the exact identity

$$X = \ell(B_0 + U - p) = \frac{B_0 + U - p}{\sqrt{pq+s}}, \qquad pq + s = \ell^{-2}, \qquad \mathbb{E}(U \mid B_0) = 0. \tag{4}$$

Indeed the conditional means are $-p\ell$ and $q\ell$, and the variance decomposition gives $1 = pq\ell^2 + \theta$. Thus the normalized count-coordinate variable is exactly $X$, and $H$ remains its original raw threshold. The small-noise proof chooses an anchor with $u \in [-3/10, 7/10]$; the interpolation proof chooses the floor anchor, with $u \in [0, 1)$. These are separate choices for separate estimates. The signed moments are $\gamma = \mathbb{E}X^3$ and $\beta = \mathbb{E}[X|X|]$.

The main text retains the variational, geometric, localization, and comparison arguments. The appendices give the full Fourier bound, constant enclosures, the identity without contact, and reproducibility instructions. The complete localization theorem is stated before its technical derivation; all its stronger auxiliary conclusions are retained.

## 2 Fixed-target extremizers and contact

We optimize the excess above a fixed candidate constant. This gives a variational problem whose coefficient is known before any acquisition step. The positive excess is retained through the contact calculation and the conserved identity; it becomes a nonnegative term in the final resource. No upper bound for the maximizing normalized score is needed. Throughout, sums have independent identically distributed summands with mean zero, variance one and finite third absolute moment $\rho$. Fix $n \geq 2$, $c > 0$, and a real raw sum threshold $H$. Set

$$z = H/\sqrt{n}, \quad F = \mathbb{P}(S_n \leq H), \quad a = \sqrt{n}\{F - \Phi(z)\} - c\rho,$$
$$\varphi = \phi(z), \quad \beta = \mathbb{E}[X|X|], \quad \gamma = \mathbb{E}X^3.$$

In particular $a$ is not divided by $\rho$.

### 2.1 Attainment, contamination and the entire closed support

**Proposition 2.1.** *For each fixed $n$ and $c > 0$, the supremum of $a$ over these laws and thresholds is attained, and the closed-CDF discrepancy at every attaining pair is positive. At every maximizing pair,*

$$I(y) = n^{3/2}\{h(y) - F\} + ny\varphi + \tfrac{1}{2}H\varphi(y^2 - 1) + \tfrac{3}{2}c\rho(y^2 - 1) - c\{|y|^3 - \rho - 3y\beta\} \leq 0, \qquad h(y) = \mathbb{P}(S_{n-1} \leq H - y), \tag{5}$$

*with equality on the whole closed support. That support is compact. Neither statement assumes $a \geq 0$ or $H \geq 0$.*

*Proof.* Write $A$ for the supremum. Since $\rho \geq 1$ and the discrepancy is at most one, $A \leq \sqrt{n}-c$. Symmetric Bernoulli evaluated at $H = n$ gives $A \geq \sqrt{n}(1-\Phi(\sqrt{n}))-c > -c$. For a maximizing sequence, $a_j \to A$ and $c\rho_j \leq \sqrt{n} - a_j$ give a common third-moment ceiling. The discrepancies satisfy $\sqrt{n}(F_j - \Phi(H_j/\sqrt{n})) = a_j + c\rho_j \geq a_j + c > \eta$ eventually for some $\eta > 0$. Chebyshev bounds the negative threshold tail by $n/H_j^2$; the positive discrepancy is at most the normal tail, also at most $n/H_j^2$. Thus $H_j$ is bounded.

Tightness gives a subsequence $P_j \Rightarrow P$, $H_j \to H$ and $\rho_j \to b$. The common third-moment ceiling $M$ gives tail bounds $\int_{|x|>K} x^2 dP_j \leq M/K$ and $\int_{|x|>K} |x| dP_j \leq M/K^2$. Bounded continuous truncations therefore preserve mean and variance. Lower semicontinuity gives $1 \leq \rho(P) \leq b$. For fixed $n$, the sum laws converge weakly. Applying the closed-set inequality first at $H + t$ and then letting $t \downarrow 0$ gives $\limsup F_j(H_j) \leq F_P(H)$, including an atom at $H$. Consequently

$$a(P, H) \geq A + c\{b - \rho(P)\} \geq A.$$

Maximality forces equality and $\rho(P) = b$. Convergence of third moments is a conclusion here; it was not inferred from bounded third moments. The discrepancy remains positive since $A + c\rho \geq A + c > 0$.

The contact inequality is a first-variation condition. To obtain it, contaminate the raw marginal by $(1 - \epsilon)P + \epsilon\delta_y$. Its mean is $\epsilon y$, variance $v_\epsilon = 1 + \epsilon(y^2 - 1) - \epsilon^2 y^2$, and centered third absolute moment has derivative $|y|^3 - \rho - 3y\beta$ at zero. The last assertion follows by dominated differentiation of $|x - \epsilon y|^3$, with bound $3|y|(|x| + |y|)^2$. Evaluate the standardized law at the standardized image of the fixed raw event $S_n \leq H$. The raw event probability is a polynomial in $\epsilon$, with derivative $n(h(y) - F)$ even at an atom collision. The normal argument is $(H - n\epsilon y)/\sqrt{nv_\epsilon}$, whose derivative is $-\sqrt{n}y - H(y^2 - 1)/(2\sqrt{n})$. The one-sided derivative of the excess is exactly (5); maximality makes it nonpositive.

Its integral against $P$ is zero. All terms are integrable. The closed-event map $h$ is upper semicontinuous; at an atom it takes the larger one-sided value. Thus $I$ is upper semicontinuous. Strict negativity at a closed-support point would hold with a uniform negative margin on a neighborhood of positive $P$-mass, contradicting its zero integral. Finally $I(y) \leq -c|y|^3 + C_2 y^2 + C_1|y| + C_0$ for constants depending on this pair. It is strictly negative on both sufficiently distant rays, proving compact support. No higher moments were used before this step. □

## 2.2 Reflection, equality, and the first unresolved index

The sign of the attained maximum detects failure of a strict row bound, including equality. If $C_n$ denotes the supremum of the normalized absolute discrepancy, reflection gives, including an atom at $H$,

$$\mathbb{P}(-S_n \leq -H) - \Phi(-z) = \Phi(z) - \mathbb{P}(S_n < H) \geq \Phi(z) - F.$$

Thus each negative closed-CDF error is bounded by a positive closed-CDF error of the reflected law; equality at an atom is not assumed. Consequently $C_n$ is also the supremum of positive closed scores $r$. Since $a = \rho(r - c)$, $C_n < c$ gives $A \leq C_n - c < 0$, and $C_n > c$ gives $A > 0$. If $C_n = c$, choose $r_j \uparrow c$ with $r_j \geq c/2$. Then $\rho_j \leq 2\sqrt{n}/c$, so $\rho_j(r_j - c) \to 0$ and $A = 0$. Hence $C_n \geq c$ supplies an attained pair with $a \geq 0$.

**Corollary 2.2** (First failure of a strict bound)**.** *Assume a genuine finite base $C_j \leq c$ for $1 \leq j < N$. If a strict bound $C_n < c$ fails at any index $n \geq N$, there is a least such index, an attained fixed-target pair with $a \geq 0$, and true predecessor bounds $C_j \leq c$ for every $j < n$. The bounds are strict for $N \leq j < n$.*

*Proof.* The nonempty set of failing indices at least $N$ has a least member. The preceding equivalence supplies its maximizing pair. Indices below $N$ use the finite base; the other earlier indices use minimality. □

Thus an argument excluding every such pair for $n \geq N$ would give strict bounds above the finite base, including exclusion of equality. The subsequent entry argument uses only $C_{n-1} \leq c$; it does not use an upper bound for the actual score $c + a/\rho$. The finite base and the eventual exclusion remain separate obligations.

Equivalently, $\Delta = Q - h \geq 0$ everywhere and $\Delta = 0$ on the closed support, where

$$Q(y) = F + \frac{H\varphi + c\rho}{2n^{3/2}} - \frac{n\varphi + 3c\beta}{n^{3/2}}y - \frac{H\varphi + 3c\rho}{2n^{3/2}}y^2 + \frac{c}{n^{3/2}}|y|^3. \tag{6}$$

### 2.3 Two elementary moment domains

The moment inequalities below are used before any finite-dimensional relaxation. They hold for the actual marginal, without a bounded-support assumption. In particular, they are unaffected by adding a very small atom far from the origin.

**Theorem 2.3** (Signed and absolute third moments)**.** *If* $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, $\rho = \mathbb{E}|X|^3 < \infty$, *and* $\gamma = \mathbb{E}X^3$, *then*

$$\rho \geq \frac{\gamma^2 + 2}{\sqrt{\gamma^2 + 4}}.$$

This is the sharp moment relation of Shevtsova [15]; in the cited arXiv version its equivalent form is Theorem 2. We include a factorization proof to specify the domain used by the finite certificates.

*Proof.* Set $s = \sqrt{\gamma^2 + 4}$, $a = (s - \gamma)/2$, $b = (s + \gamma)/2$ and $c = \gamma(\gamma^2 + 6)/s^3$. Then $a, b > 0$, $ab = 1$ and $a + b = s$. Define

$$w(y) = |y|^3 - cy^3 - \frac{12y^2}{s^3} + \frac{6\gamma y}{s^3} + \frac{4}{s^3}.$$

For $y, t \geq 0$, direct multiplication gives

$$w(y) = \frac{2a^2}{s^3}(y - b)^2\{2 + (a + 3b)y\}, \qquad w(-t) = \frac{2b^2}{s^3}(t - a)^2\{2 + (3a + b)t\}.$$

Thus $w \geq 0$ on the whole real line. Taking expectations and using the first three signed moments gives

$$0 \leq \mathbb{E}w(X) = \rho - c\gamma - 8/s^3 = \rho - (\gamma^2 + 2)/s.$$

All terms are integrable under the stated third-moment hypothesis. □

**Lemma 2.4** (The signed second moment)**.** *With the same normalization and* $\beta = \mathbb{E}[X|X|]$,

$$|\beta| < 1, \qquad \rho \geq \frac{1 + \beta^2}{\sqrt{1 - \beta^2}}.$$

*Proof.* Both signs have positive probability. Put $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$ and $a_\pm = \mathbb{E}X_\pm^2 = (1 \pm \beta)/2 > 0$. If $p_\pm = \mathbb{P}(\pm X > 0)$, Cauchy's inequality gives $m^2 \leq p_\pm a_\pm$. Since $p_+ + p_- \leq 1$, this implies

$$m^2(1/a_+ + 1/a_-) \leq 1, \qquad 2m \leq \sqrt{1 - \beta^2}.$$

Another application of Cauchy's inequality, to $X_\pm^{1/2}$ and $X_\pm^{3/2}$, gives $m\mathbb{E}X_\pm^3 \geq a_\pm^2$. Summing yields $m\rho \geq (1 + \beta^2)/2$, which proves the claim. A possible zero atom enters only through $p_+ + p_- \leq 1$ and causes no exception. □

# 3 The two-point skeleton and its moment defect

Every centered law can be separated into its positive and nonpositive parts. Replacing each part by its conditional mean gives a two-point law; the remaining conditional fluctuations measure its departure from that law. We first make this decomposition exact, then identify a moment defect that vanishes at Esseen's extremal distribution.

## 3.1 The sign split and its units

Fix $P \in \mathcal{P}$ and put

$$p = \mathbb{P}(X > 0), \quad q = 1 - p, \quad v = pq, \quad m = \mathbb{E}X_+ = \mathbb{E}X_- > 0, \quad B_0 = \mathbf{1}_{\{X>0\}}.$$

Both $p, q$ are positive; a possible atom at zero is assigned to the nonpositive class. Define $V = \mathbb{E}[X \mid B_0]$ and $W = X - V$, so that

$$V = -m/q \quad (B_0 = 0), \qquad V = m/p \quad (B_0 = 1), \qquad \theta = \mathbb{E}W^2 = 1 - m^2/v < 1.$$

Write $\theta_- = \mathbb{E}[W^2; B_0 = 0]$ and $\theta_+ = \mathbb{E}[W^2; B_0 = 1]$. The distance between the two conditional means is

$$\ell = m/v = \sqrt{(1-\theta)/v} > 0.$$

Set $U = W/\ell$ and $Y = B_0 + U$. Then

$$X = \ell(Y - p), \qquad \mathbb{E}[U \mid B_0] = 0. \tag{7}$$

Let $U_0, U_1$ have the two conditional laws, and put

$$w_i = \mathbb{E}U_i^2, \quad \mu_i = \mathbb{E}U_i^3, \quad s = qw_0 + pw_1 = \frac{v\theta}{1-\theta}, \quad \lambda = ns, \quad \ell = (v+s)^{-1/2}.$$

The moments $\mu_i$ are finite because $\rho < \infty$. Higher moments will be used only after bounded support is known: for extremizers this follows from Proposition 2.1, and for the local comparison it is an explicit hypothesis.

If $s = 0$, both nonnegative conditional variances vanish, so $U_0 = U_1 = 0$ almost surely. This is exactly the standardized Bernoulli law. If $s > 0$, the parameter $\alpha = pw_1/s$ lies in $[0, 1]$ and gives $w_0 = s(1-\alpha)/q$, $w_1 = s\alpha/p$.

The signs of $X$ are still those defining the two classes, so expansion of the actual conditional moments gives

$$\begin{aligned}
\rho/\ell^3 &= v(1-2v) + 3v(w_0 + w_1) - q\mu_0 + p\mu_1, \\
\gamma/\ell^3 &= v(q-p) + 3v(w_1 - w_0) + q\mu_0 + p\mu_1, \\
\beta/\ell^2 &= v(q-p) + pw_1 - qw_0.
\end{aligned} \tag{8}$$

For example, on the nonpositive class $|X|^3 = \ell^3(p - U_0)^3$; on the positive class it is $\ell^3(q + U_1)^3$. Conditional centering removes the linear terms and gives the first identity; the other two follow in the same way. The three identities share the same pair $(\mu_0, \mu_1)$.

## 3.2 A nonnegative moment defect

The constant $3 + \sqrt{10}$ has a simple geometric explanation. The signed third moment and the first absolute moment satisfy a circle inequality; the lengths of the support gaps are controlled by the same first absolute moment. Neither fact requires bounded support.

Define

$$\tau(t) = \mathbb{E}[X\mathbf{1}_{\{X\geq t\}}], \qquad \sigma = \sqrt{10}.$$

For $t < 0$, centering gives $\tau(t) = -\mathbb{E}[X\mathbf{1}_{\{X<t\}}] \geq 0$; for $t \geq 0$ nonnegativity is immediate. Signwise Tonelli integration gives

$$\int_{\mathbb{R}} \tau(t)\,dt = 1, \qquad 2\int_{\mathbb{R}} |t|\tau(t)\,dt = \rho. \tag{9}$$

On each bounded complementary interval $(a, b)$ of the closed support, $\tau$ is constant in the interior, with value $\tau_{a,b}$. Put

$$J = \sum_{(a,b)} \tau_{a,b}(b-a)^2, \qquad D = (3+\sigma)\rho - \gamma - 3J. \tag{10}$$

There are at most countably many gaps; the next proof shows that the nonnegative sum is finite. Endpoint atoms do not affect its integrals.

**Lemma 3.1** (A moment circle and the gap inequality)**.** *For every $P \in \mathcal{P}$,*

$$\gamma^2 + (2m)^2 \leq \rho^2, \qquad J \leq \rho + 2m, \qquad \gamma + tJ \leq \big(t + \sqrt{1+t^2}\big)\rho \quad (t > 0). \tag{11}$$

*The last constant is sharp. Equality holds exactly at the standardized Bernoulli law with positive mass $p_t = (1 + t - \sqrt{1+t^2})/2$. In particular, $D \geq 0$. If zero belongs to the support, then $D \geq (\sigma - 1)\rho$; hence $D < (\sigma - 1)\rho$ implies a central support gap $(-A, B)$ with $A, B > 0$.*

*Proof.* Write $a_\pm = \mathbb{E}X_\pm^2$ and $b_\pm = \mathbb{E}X_\pm^3$, so $a_+ + a_- = 1$, $b_+ + b_- = \rho$ and $b_+ - b_- = \gamma$. Cauchy–Schwarz on each sign class gives $a_\pm^2 \leq m b_\pm$. It also gives $m^2 \leq p a_+$ and $m^2 \leq q a_-$, where an atom at zero is included in $q$. Thus $m^2(1/a_+ + 1/a_-) \leq 1$, or $m^2 \leq a_+ a_-$. Consequently $b_+ b_- \geq (a_+ a_-)^2/m^2 \geq m^2$, proving the circle inequality since $\rho^2 - \gamma^2 = 4b_+b_-$.

Every gap on one side of zero satisfies $(b-a)^2 \leq 2\int_a^b |x|\,dx$. If a central gap $(-A, B)$ exists, its weight is $m$ and its excess over this integral is $2mAB$. Moreover $0 \leq \mathbb{E}[(X+A)(X-B)] = 1 - AB$. Summing the gap bounds by Tonelli proves $J \leq \rho + 2m$. If zero is in the support there is no central gap, so $J \leq \rho$. Finally Cauchy–Schwarz in $\mathbb{R}^2$ gives $\gamma + 2tm \leq \sqrt{1+t^2}\rho$, proving the family and the assertions about $D$.

Equality in the circle forces equality in the two signwise Cauchy bounds: the positive and negative parts are each concentrated at one nonzero value. Equality in $m^2 \leq a_+ a_-$ excludes mass at zero. For such a law, with positive mass $p$ and $q = 1 - p$, one has $J = \rho + 2m$. Equality in the last Cauchy bound requires $\gamma = 2m/t$, equivalently $t(1-2p) = 2p(1-p)$. Its unique solution in $(0, 1)$ is the stated $p_t$. □

**Corollary 3.2** (The zero-defect law)**.** *For $P \in \mathcal{P}$, $D = 0$ if and only if*

$$X = (B_0 - p_*)/\sqrt{p_* q_*}, \qquad p_* = (4 - \sqrt{10})/2, \quad q_* = 1 - p_*, \quad B_0 \sim \text{Bernoulli}(p_*).$$

*Proof.* Apply the equality case of Lemma 3.1 with $t = 3$. □

In particular the defect separates two losses:

$$D = \underbrace{\sigma\rho - \gamma - 6m}_{\text{moment loss}} + 3\underbrace{(\rho + 2m - J)}_{\text{gap loss}},$$

each nonnegative. The point $(\gamma/\rho, 2m/\rho)$ lies in the upper unit disk. The line $x + 3y = \sqrt{10}$ supports that disk at $(1/\sqrt{10}, 3/\sqrt{10})$, explaining the coefficient in the moment loss. For a

standardized Bernoulli law with positive mass $p$, direct calculation gives

$$J = \frac{1}{\sqrt{pq}}, \qquad \rho_B(p) = \frac{1-2pq}{\sqrt{pq}},$$
$$\gamma_B(p) = \frac{q-p}{\sqrt{pq}}, \qquad D_B(p) = \frac{(6+2\sigma)(p-p_*)^2}{\sqrt{pq}}. \tag{12}$$

Indeed its zero-bias density is $\sqrt{pq}$ on the interval between its two support points. The probability displacement therefore has a quadratic cost. The equality proof above is what rules out other laws with zero defect.

For quantitative estimates we need more information than their signs. Suppose a central gap $(-A, B)$ has been established, and write

$$T_+ = \mathbb{E}X_+^3 - mB^2, \quad T_- = \mathbb{E}X_-^3 - mA^2, \qquad R_+ = \mathbb{E}X_+^2 - mB, \quad R_- = \mathbb{E}X_-^2 - mA.$$

These four quantities are nonnegative. For the outer gaps put $J_o = J - m(A+B)^2$. A positive outer gap satisfies $(b-a)^2 \le 2\int_a^b (t-B)\,dt$, so Tonelli gives

$$J_o^+ \le 2\int_B^\infty (t-B)\tau(t)\,dt = \mathbb{E}[X_+(X_+-B)^2] = T_+ - 2BR_+ \le T_+.$$

Similarly $J_o^- \le T_- - 2AR_- \le T_-$. Expanding $D$, using $\sqrt{\sigma+1}\sqrt{\sigma-1} = 3$, gives

$$D = m\{\sqrt{\sigma+1}A - \sqrt{\sigma-1}B\}^2 + (\sigma-1)T_+ + (\sigma+1)T_- \\ + 3(T_+ + T_- - J_o). \tag{13}$$

Every term is nonnegative. Cauchy–Schwarz also gives $1/(2\rho) \le m \le 1/2$: indeed $1 \le \rho\mathbb{E}|X| = 2m\rho$ and $\mathbb{E}|X| \le 1$. Since $\sigma - 1 > 2$, the decomposition yields $T_+ + T_- \le D/2$. If $D < 2$, then

$$m(A^2+B^2) = \rho - T_+ - T_- \ge 1 - D/2, \qquad A + B \ge \sqrt{2-D}.$$

### 3.3 One joint budget for endpoint motion and noise

Let $\bar{A} = m/q = \ell p$, $\bar{B} = m/p = \ell q$ be the conditional mean magnitudes. For a central support gap $(-A, B)$ define

$$\Delta_A = \bar{A} - A, \quad \Delta_B = \bar{B} - B, \qquad a_0 = \Delta_A/\ell, \quad a_1 = \Delta_B/\ell, \quad d = a_0 + a_1.$$

These quantities are nonnegative, $d < 1$, and $U_0 \le a_0$, $U_1 \ge -a_1$. The physical displacement $\Delta_A + \Delta_B$ is $\ell d$; the displacement $d$ is in count units. Put $w = 1 - \theta = pq\ell^2$ and

$$\mathcal{Q}(p) = (\sigma+1)p^2 + (\sigma-1)q^2 - 6pq = (6+2\sigma)(p-p_*)^2.$$

**Lemma 3.3** (Joint defect budget)**.** *For every $P \in \mathcal{P}$ whose closed support has a central gap $(-A, B)$ with $A, B > 0$,*

$$D \ge \ell w\mathcal{Q}(p) + 6\ell w\{d(1-d) + a_0a_1\} \\ + \mathbb{E}[\{(\sigma-1)(X+2\bar{B}) + 6B\}W^2; X > 0] \\ + \mathbb{E}[\{(\sigma+1)(-X+2\bar{A}) + 6A\}W^2; X < 0]. \tag{14}$$

*All terms on the right are nonnegative. They measure, respectively, probability displacement, inward motion of the two gap endpoints, and conditional variance at its actual support locations.*

*Proof.* Write $M_{j,\pm} = \mathbb{E}X_{\pm}^{j}$. The outer-gap bounds proved above give

$$J_o^+ \le M_{3,+} - 2BM_{2,+} + B^2 m, \qquad J_o^- \le M_{3,-} - 2AM_{2,-} + A^2 m.$$

Adding the central term $m(A+B)^2$ and substituting in $D$ yields

$$\begin{aligned} D \ge (\sigma-1)M_{3,+} + (\sigma+1)M_{3,-} + 6BM_{2,+} + 6AM_{2,-} \\ - 6m(A^2 + AB + B^2). \end{aligned}$$

Conditional centering gives the exact identities

$$M_{2,+} = m\bar{B} + \mathbb{E}[W^2; X > 0], \qquad M_{3,+} = m\bar{B}^2 + \mathbb{E}[(X + 2\bar{B})W^2; X > 0],$$

and the analogous identities with $-X, \bar{A}$ on the negative class. After substituting $A = \ell(p - a_0)$ and $B = \ell(q - a_1)$, the deterministic terms are

$$\ell w\{\mathcal{Q}(p) + 6[d(1-d) + a_0 a_1]\};$$

the remaining terms are exactly the two expectations in (14). The gap sums are nonnegative, so monotone convergence justifies summing countably many gaps. Each expectation is integrable under the finite third-moment assumption; endpoint atoms do not affect the Lebesgue integrals. □

**Corollary 3.4** (Simultaneous gap inequality)**.** *For this central gap,*

$$\begin{aligned} D \ge \frac{(6+2\sigma)(1-\theta)^{3/2}}{\sqrt{pq}}(p - p_*)^2 + 6m\{(A+B)(\Delta_A + \Delta_B) + \Delta_A\Delta_B\} \\ + 3(\sigma+1)B\theta_+ + 3(\sigma+3)A\theta_-. \end{aligned} \tag{15}$$

*Proof.* On the positive support $X \ge B$ and $\bar{B} \ge B$, so the positive kernel in (14) is at least $3(\sigma+1)B$. On the negative support $-X \ge A$ and $\bar{A} \ge A$, the corresponding kernel is at least $3(\sigma+3)A$. Finally $m = pq\ell$ and $A + B = \ell(1-d)$ identify the two deterministic terms with those in (15). □

# 4 Conservation and control of the defect

The equality case in Section 3 identifies the extremal Bernoulli law. We now connect this geometry to the excess: contact gives an identity in which $D$ appears with a positive coefficient. Bounding its remainder will then bound the distance from that equality case.

Retain the definitions of $a, Q, h$ and $\Delta = Q - h$ from Section 2. For now let $(P, H)$ be an attained pair at any $c > 0$, so the law has compact support $K$ and $\Delta = 0$ on $K$.

## 4.1 Zero-bias integration and the conservation identity

Put $\nu = \mathcal{L}(H - S_{n-1})$ and $\tau(t) = \mathbb{E}[X\mathbf{1}_{X \ge t}]$. Then $\tau \ge 0$, $\int \tau = 1$. Over the bounded complementary intervals $(u, v)$ of the closed support $K$, define

$$\begin{aligned} J &= \sum \tau_{u,v}(v-u)^2, & J_1 &= \sum \tau_{u,v} \int_u^v (t-u)t\,dt, \\ J_2 &= \sum \tau_{u,v} \int_u^v (t-u)t|t|\,dt, & L &= \sum \tau_{u,v} \int_{[u,v)} (t-u)\,d\nu(t). \end{aligned}$$

Here $\tau_{u,v}$ is the constant interior value; $J$ and $\tau$ agree with the definitions in Section 3. The leakage $L$ is nonnegative. It uses $[u, v)$: the right endpoint belongs to $h(v)$, whereas

an atom at the left endpoint has zero weight. If $K \subset [-R, R]$, then $J \le 2R$, $|J_1| \le RJ/2$, $|J_2| \le R^2 J/2$, and $0 \le L \le 2R$, so all gap sums converge.

The remainder has four parts: a constant moment and gap term, a linear threshold term, a quadratic threshold term, and the Fourier error $E_n$. Specifically, let $\varphi = \phi(H/\sqrt{n})$, $M_4 = \mathbb{E}X^4$, $M_5 = \mathbb{E}|X|^5$, and put

$$K_q = M_5/4 - \rho M_4/2 - 3\beta\gamma/2 + \rho/2, \qquad K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2,$$

$$\Omega = J_1 + (M_4 - 3)/6, \quad \varepsilon = \mathbb{E}q_z(S_n/\sqrt{n}) - \mathbb{E}q_z(Z_0) - 2\gamma\varphi/(3\sqrt{n}), \quad q_z(x) = (x - z)|x - z|,$$

where $z = H/\sqrt{n}$ and $Z_0$ is standard normal. Define $E_n = -n^{3/2}\varepsilon/2$ and

$$\mathcal{R}_c = c(3\beta J/2 + 3\rho J_1 - 3J_2 - K_q) + H(cK_1 + \varphi\Omega) - H^2(c\rho + \gamma\varphi/2) + E_n. \tag{16}$$

**Theorem 4.1** (Conservation at contact)**.** *At an attained pair with $c = C_{\mathrm{E}}$, put $e = \exp(-H^2/(2n))$. Then*

$$\mathcal{R}_c = (n + H^2)a + nc\rho(1 - e) + \frac{n\phi(0)e}{6}D + n^{3/2}L. \tag{17}$$

*More generally, at an attained pair for any $c > 0$,*

$$(n + H^2)a + nc\rho - \frac{n\varphi}{6}(\gamma + 3J) + n^{3/2}L = \mathcal{R}_c. \tag{18}$$

In (17), the four terms retain the excess, the loss from a nonzero normal threshold, the departure from the zero-defect law, and the nonnegative leakage of the reflected predecessor sum through support gaps. When $a \ge 0$, all four are nonnegative. The proof must therefore bound $\mathcal{R}_c$ from above without assuming that the excess is already negative. The coefficient of the unnormalized excess is $n + H^2$.

*Proof.* We evaluate $\mathcal{A} = \int \tau(Q - h)$ in two ways. Gapwise integration uses contact at both endpoints. Zero-bias integration converts the polynomial part to moments and the conditional CDF part to a truncated moment of the sum. Equating these expressions gives the identity. Since $h(t) = \nu([t, \infty))$, one has $h(t) = h(v) + \nu([t, v))$ inside a gap. Thus bounded Fubini gives

$$\int_u^v \Delta(t)dt = (v - u)\Delta(v) + \int_u^v (t - u)(-Q'(t))dt - \int_{[u,v)} (t - u)d\nu(t).$$

An atom at $v$ belongs to $h(v)$ and is not in the last integral; an atom at $u$ is included but multiplied by zero. Because $\Delta = 0$ on the entire closed support, the endpoint terms and the integral over $K$ vanish. Summing gives

$$\mathcal{A} = \frac{\varphi J}{2\sqrt{n}} + \frac{3c\beta J/2 + (H\varphi + 3c\rho)J_1 - 3cJ_2}{n^{3/2}} - L. \tag{19}$$

For the second evaluation, signwise Fubini proves $\int \tau g' = \mathbb{E}[Xg(X)]$ for absolutely continuous $g$ with bounded derivative on the relevant compact range. In particular

$$\int t\tau = \gamma/2, \quad \int t^2\tau = M_4/3, \quad \int |t|^3\tau = M_5/4.$$

Apply that same identity conditionally to $g_s(x) = \min(s + x, H)$ and then use exchangeability. The choice of derivative at its single corner is immaterial under the density $\tau$. It follows directly that

$$\int \tau(t)h(t)dt = \mathbb{E}[X_n \min(S_n, H)] = \frac{1}{n}\mathbb{E}[S_n \min(S_n, H)] = T,$$

where $T = \mathbb{E}[(Z^2 - zZ)\mathbf{1}_{Z\le z}]$, $Z = S_n/\sqrt{n}$. The last expression vanishes at an endpoint atom. Integrating (6) against $\tau$ therefore yields

$$\mathcal{A} = F - \frac{\gamma\varphi}{2\sqrt{n}} + \frac{H\varphi(3 - M_4)/6 + cK_q}{n^{3/2}} - T. \tag{20}$$

Set $M = \mathbb{E}[Z\mathbf{1}_{Z\le z}]$. Since $\mathbb{E}[Xh(X)] = M/\sqrt{n}$, integration of $XQ(X)$ against $P$ gives

$$M + \varphi = -\frac{\gamma H\varphi}{2n} + \frac{cK_1}{n}. \tag{21}$$

The pointwise identity $q_z(x) = (x - z)^2 - 2(x - z)^2\mathbf{1}_{x\le z}$, with the same identity for a normal variable, gives

$$T - \Phi(z) = -\varepsilon/2 - \gamma\varphi/(3\sqrt{n}) + z(M + \varphi) - z^2(F - \Phi(z)).$$

Inserting (21) and $\sqrt{n}(F - \Phi(z)) = c\rho + a$ into (20) gives

$$\begin{aligned}\sqrt{n}\mathcal{A} &= c\rho + a - \gamma\varphi/6 + P_0/n + \sqrt{n}\varepsilon/2,\\ P_0 &= H^2(c\rho + a + \gamma\varphi/2) - H\{cK_1 + \varphi(M_4 - 3)/6\} + cK_q.\end{aligned}$$

Equate this with (19) multiplied by $\sqrt{n}$, then multiply by $n$ and collect terms. This is (18). Substituting $c = C_{\mathrm{E}} = (3 + \sqrt{10})\phi(0)/6$, $\varphi = \phi(0)e$, and $\gamma + 3J = (3 + \sqrt{10})\rho - D$ proves (17). □

The compensated Fourier representation of this same $E_n$ is proved in Appendix A. Its comparator has transform $e^{-nt^2/2}(1 - in\gamma t^3/6)$. Appendix C retains the stronger identity for arbitrary compactly supported laws, including the residuals that disappear here by contact.

### 4.2 Why the remainder bound must come first

By Lemma 3.1,

$$D = 3(\rho + 2m - J) + (\sqrt{10}\rho - \gamma - 6m) \ge 0. \tag{22}$$

Specialize from now on to $c = C_{\mathrm{E}}$ and $a \ge 0$. A first bound for $\mathcal{R}_c$ proves a positivity condition; only then can the threshold term improve the bound for the nonnegative expression below.

**Corollary 4.2** (From a remainder bound to a defect bound)**.** *At such a contact pair with $a \ge 0$, assume $n \ge N \ge 2$, $\rho \ge l > 0$, $0 < c_- \le c$, and a proved raw bound $\mathcal{R}_c \le T < Nc_-l$. A negative $T$ excludes the pair. Otherwise*

$$\begin{gathered}\gamma + 3J > 0, \qquad e \ge 1 - T/(Nc_-l) > 0,\\ \mathcal{U} := (n + H^2)a + n\phi(0)D/6 + n^{3/2}L \le \mathcal{R}_c,\\ \mathcal{U} \le \mathcal{R}_c - \tfrac{1}{2}(c_-l - T/N)H^2.\end{gathered}$$

*No condition $D < 2$ or own-score upper bound is required.*

*Proof.* Set $M_c = c\rho - \phi(0)D/6 = \phi(0)(\gamma + 3J)/6$. The contact identity gives

$$eM_c = (1 + H^2/n)a + c\rho + \sqrt{n}L - \mathcal{R}_c/n \ge c\rho - T/n > 0.$$

Thus $M_c > 0$ and, since $D \ge 0$, $M_c \le c\rho$. The density bound follows. Regrouping (17) gives $\mathcal{R}_c = \mathcal{U} + n(1 - e)M_c$, proving the unweighted bound. Finally, put $w = H^2/(2n)$, so $e = e^{-w}$. Then

$$n(1 - e)M_c = n(e^w - 1)eM_c \ge \frac{H^2}{2}(c\rho - T/n) \ge \frac{H^2}{2}(c_-l - T/N).$$

This is a bound on $\mathcal{U}$, not an improved bound on the raw remainder. □

Every conclusion holds at each actual index and raw threshold. Common bounds on a closed moment cell make the result uniform even when the maximizing law varies with $n$. There is no exchange of a limit with contact, support gaps, moments, or an integral. In particular, the order of use is essential: a raw ceiling first proves $M_c > 0$ and a density floor; only then may the Gaussian phase term be subtracted to improve $\mathcal{U}$. A smaller resource ceiling is not a replacement for the raw ceiling in that first step.

# 5 Quantitative localization

The zero-defect law is now known. We need a quantitative statement strong enough to control a possible extremizer. Conservation reduces this to an upper bound for its remainder. Contact bounds the support, which permits higher-moment and Fourier estimates. Iterating these estimates gives the following theorem; its two principal outputs for the next section are $nD \le 32.20665378$ and $|H| \le 1.83605680$.

**Theorem 5.1** (Fixed-target coarse localization)**.** *Let $n \ge 42$ be an actual index and let an attained fixed-target pair at $C_{\mathrm{E}}$ have $a \ge 0$. Assume $C_{n-1} \le C_{\mathrm{E}}$, the external structural inequality* (51)*, and Conditions D.3 and D.5. Then the complete probability cover specified in Appendix D applies to that pair, and simultaneously*

$$\mathcal{R}_c \le 2.1995174, \qquad nD \le 32.20665378, \qquad D \le 0.76682509,$$
$$\rho \le 71527/48000, \qquad |\beta| \le 0.50549963, \qquad |H| \le 1.83605680,$$
$$\operatorname{supp} P \subset (-157/50, 157/50).$$

The full analytical proof and all finite conditions are in Appendix D. The Fourier bounds in Appendix A apply uniformly to every actual $n \ge 42$, even when the law and threshold vary with $n$. Evaluating their scalar majorants at the floor $N = 42$ is therefore sufficient; it does not set the actual sample size equal to 42.

## 5.1 How the bounds improve

The steps have distinct purposes:

1. The structural inequality bounds $\rho$. Contact and the sharp predecessor then bound the entire support, uniformly in $H$.

2. Bounded support supplies higher-moment bounds and an event for $X - X'$ under two independent copies of the original law. This event controls the nonzero-frequency peaks in the Fourier integral.

3. Two coarse passes bound the remainder and then the defect, proving that zero is outside the support. Thus the original law has an actual central gap.

4. Four signed-moment passes retain the gap geometry and the signs of the moment terms. They improve the same-law bounds and give the outputs of the theorem.

Choose rational constants $0 < c_- < C_{\mathrm{E}} < c_+$ and $0 < \phi_l < \phi(0)$; their explicit values and proofs are given in Appendices D and B. For example, suppose the coefficient bounds on a cell with $\rho \ge l$ give, for $v = c_-, c_+$,

$$\mathcal{R}_c \le \max_v \{B_v + F_v|H| - Q_v H^2\}, \qquad Q_v > 0.$$

Completing the square bounds this by $\max_v(B_v + F_v^2/(4Q_v))$. An outward rational upper bound $T$ with $T < Nc_-l$ first establishes the positivity needed in Corollary 4.2. Only then

do the larger curvatures $Q_v + (c_- l - T/N)/2$ give a bound $U$ for $\mathcal{U}$, and hence $nD \le 6U/\phi_l$. This explains both the update and its order: $U$ bounds a nonnegative expression, not the original remainder.

### 5.2 Why the finite verification covers all witnesses

**Proposition 5.2** (Soundness of the finite localization)**.** *Suppose the enclosure and exclusion rules proved in Appendices A and D are evaluated in the stated directions, every required inequality passes, and every closed-cover incidence is checked. Then the resulting bounds hold simultaneously for every attained nonnegative-excess pair in the initial domain at every actual $n \ge 42$ with the sharp predecessor.*

*Proof.* The initial 24 closed $\rho$-cells cover the moment range supplied by Theorem D.1. Fix one possible witness in one of these cells. Every subdivision covers its parent. Intersecting necessary moment inequalities keeps this witness; the original $\beta$-interval is restored separately for each $\gamma$-slice. A child is discarded only by an incompatible necessary inequality. In particular, a positive numerator with zero denominator is treated using the original moment matrix, and a zero-over-zero correction is dropped.

For each surviving child, a support bound is proved before higher moments are used. The actual-event and Fourier estimates then bound the remainder, and the ordered update bounds the nonnegative expression in conservation. Taking maxima of upper bounds and minima of lower bounds can enlarge the possible set but cannot remove the fixed witness. A parent disappears only if all descendants have been excluded. Induction over the two coarse and four signed passes preserves this invariant.

The final closed probability panels cover the surviving probability intervals. At a shared endpoint the cell contributes to every incident panel. Pooling the simultaneous bounds therefore retains the conclusions for the same law. Finally, the analytical all-index comparisons in the Fourier appendix justify each floor evaluation at every actual $n \ge 42$. The numbers of processed boxes are accounting checks, not substitutes for this coverage argument. □

## 6 From small defect to a common support region

The entry calculation supplies the weighted defect and raw-threshold bounds. The defect bound first locates the sign probability and controls the inward endpoints and conditional variance. Two applications of contact then bound the entire conditional support. This last step uses the sharp predecessor bound; the preceding geometric estimates do not. Throughout this section set

$$\sigma = \sqrt{10}, \qquad p_* = (4 - \sigma)/2, \qquad B_* = 32.20665378.$$

The law, support gaps and sign split are those of Section 3; in particular $D$ is the actual gap defect, not a moment surrogate. Write the central gap, once acquired, as $(-A_-, A_+)$, so that its endpoints are not confused with the sign indicator $B_0 = \mathbf{1}_{\{X>0\}}$.

**Proposition 6.1** (Direct geometry)**.** *Let $X$ be compactly supported, centered and of variance one. If $n \ge 700$ and $nD \le B_*$, its closed support has a gap $(-A_-, A_+)$ across zero, with $A_-, A_+ > 0$. In the actual decomposition $X = \ell(B_0 + U - p)$, put*

$$a_0 = p - A_-/\ell, \quad a_1 = q - A_+/\ell, \quad d = a_0 + a_1,$$
$$\theta = \mathbb{E}\{X - \mathbb{E}(X \mid B_0)\}^2, \qquad \lambda = n\mathbb{E}U^2 = n\theta/\ell^2.$$

*Then* $a_0, a_1 \ge 0$, $\mathbb{E}(U \mid B_0) = 0$, $U_0 \le a_0$, $U_1 \ge -a_1$, *and*

$$\frac{3}{8} < p < \frac{93}{200}, \qquad \theta < .0036, \qquad d < .004, \qquad \ell > 1.996, \tag{23}$$

$$\rho < \frac{9}{8}, \qquad .066148 < \beta < .2527, \qquad \lambda < \frac{7}{20}. \tag{24}$$

*These conclusions require no contact or predecessor hypothesis.*

The estimates are obtained in the following order. Each preliminary bound supplies a hypothesis for a sharper one later in the proof.

| Input | Purpose of the output |
|---|---|
| $D < .047$ and the zero-in-support bound | Establish a central gap and nonnegative inward displacements. |
| The joint defect budget | Obtain $.36 < p < .48$, $\theta < .006$ and $\ell > 1.99$ before taking derivatives of the refined bounds. |
| The inward-motion term | Give $d < .004$. |
| Monotonicity in $d$ and $\theta$ | Give the final probability interval, $\theta < .0036$ and $\ell > 1.996$. |
| The conditional cubic costs | Control the total count variance $\lambda$, rather than only the variance of $X$. |
| Contact and the predecessor bound | Exclude the two outer support rays in Proposition 6.2. |

*Proof.* We have $D < D_0 := .047$, because $700D_0 > B_*$. If zero were in the closed support, Lemma 3.1 would give $D \ge (\sigma - 1)\rho \ge \sigma - 1 > D_0$. Thus the central gap exists; centering supplies both signs. Conditional means imply $a_j \ge 0$ and $0 \le d < 1$. Put $v = pq$ and $w = 1 - \theta = v\ell^2$, and define

$$\mathcal{Q}(p) = (\sigma + 1)p^2 + (\sigma - 1)q^2 - 6pq = (6 + 2\sigma)(p - p_*)^2.$$

We first retain all the costs of the actual conditional noises. Let

$$\int f \, d\mu_\pm = \mathbb{E}\big[(X - \mathbb{E}(X \mid B_0))^2 \mathbf{1}_{\{\pm X > 0\}} f(|X|)\big].$$

Their masses sum to $\theta$. The outer-gap estimate gives the common budget

$$D \ge \ell w \mathcal{Q} + E_0 + \int k_+ \, d\mu_+ + \int k_- \, d\mu_-, \tag{25}$$

where

$$E_0 = 6\ell w\{d(1 - d) + a_0 a_1\},$$
$$k_+(x) = (\sigma - 1)(x + 2\ell q) + 6A_+, \qquad k_-(x) = (\sigma + 1)(x + 2\ell p) + 6A_-.$$

This is Lemma 3.3 with $A = A_-$ and $B = A_+$. The measures $\mu_\pm$ are only notation for the displayed variance-weighted expectations; no new law is introduced.

Hereafter use $3.162277660 < \sigma < 3.162277661$. The weaker gap estimate $J \le \rho + 2mA_-A_+$, together with $\rho \ge 1$, $\gamma \le \rho$ and $m \le 1/2$, gives

$$A_-A_+ \ge z := \frac{3.162277660 - 1 - D_0}{3} > 0, \qquad w \ge A_-A_+ \ge z.$$

The second inequality follows from $A_- \le \ell p$, $A_+ \le \ell q$. All costs in (25) are nonnegative, so

$$D \ge \frac{w^{3/2}\mathcal{Q}}{\sqrt{v}} \ge 2z^{3/2}\mathcal{Q}.$$

Let $\mathcal{Q}_-$ replace $\sigma$ in its polynomial expression by the lower rational bound. The two exact tests

$$4z^3\mathcal{Q}_-(.36)^2 > D_0^2, \qquad 4z^3\mathcal{Q}_-(.48)^2 > D_0^2$$

exclude $p \le .36$ and $p \ge .48$, respectively: the squared distance to $p_* \in (.36, .48)$ is monotone outside that interval.

On $.36 < p < .48$, $\ell \ge 2\sqrt{z} > 1.67$, and the noise prices satisfy

$$k_+(x)/\ell \ge q\{(\sigma+5)z/w + 2(\sigma-1)\} > 5.2,$$
$$k_-(x)/\ell \ge p\{(\sigma+7)z/w + 2(\sigma+1)\} > 5.2.$$

For the first use $x \ge A_+$ and $A_+ \ge z/(\ell p)$; for the second use their negative counterparts. Since $w \le 1$, the constants follow by taking $q = .52$ and $p = .36$. Therefore $D \ge 1.67(5.2)\theta$, whence $\theta < .006$, $w > .994$ and $\ell > 1.99$. Also $A_- + A_+ \ge 2\sqrt{z}$ gives

$$d \le 1 - 2\sqrt{zv/w} < .2, \qquad v \ge .36(.64).$$

Evaluating the kernels at their lower support endpoints in (25) now yields

$$D \ge \ell\{w\mathcal{Q} + 6wd(1-d) + \theta\min(C_+ - b_+d, C_- - b_-d)\}, \tag{26}$$

where

$$C_+ = 3(\sigma+1)q, \quad C_- = 3(\sigma+3)p, \quad b_+ = \sigma+5, \quad b_- = \sigma+7.$$

Each branch increases in $d$ on the current domain, since $6(.994)(.6) - .006(3.162277661+7) > 0$; both noise prices are positive. The inward cost alone excludes $d \ge .004$, because

$$1.99(.994)6(.004)(.996) > .047.$$

After replacing $d$ by zero, the derivative in $\theta$ of either branch $\sqrt{(1-\theta)/v}\{(1-\theta)\mathcal{Q} + \theta C\}$ has the sign of $(2-3\theta)C - 3(1-\theta)\mathcal{Q}$. It is positive: $C > 6.49$, $\mathcal{Q} < .05$ and $\theta < .006$. Hence $D \ge \mathcal{Q}/\sqrt{v}$. Since $\sqrt{v} \le 1/2$, it follows that $D \ge 2\mathcal{Q}(p)$. Moreover

$$\frac{3}{8} < p_* < \frac{21}{50} = \frac{1}{2}\left(\frac{3}{8} + \frac{93}{200}\right).$$

Every $p$ outside $(3/8, 93/200)$ is therefore at least as far from $p_*$ as $3/8$, so $\mathcal{Q}(p) \ge \mathcal{Q}(3/8)$. The single rational comparison

$$\mathcal{Q}_-(3/8) = \frac{37936011}{1600000000} > \frac{47}{2000} = \frac{D_0}{2}$$

contradicts $D < D_0$ on both outside rays, including the endpoints. Thus $3/8 < p < 93/200$. Since

$$C \ge \min\{3(\sigma+1)(107/200),\ 3(\sigma+3)(3/8)\},$$

the bound $D \ge \ell\theta C$, with $\ell > 1.99$, improves $\theta < .0036$ and consequently $\ell > 1.996$.

The exact conditional cubic expansion gives

$$\rho = w^{3/2}\rho_B(p) + \int (x + 2\ell q)\, d\mu_+ + \int (x + 2\ell p)\, d\mu_-, \qquad \rho_B(p) = \frac{1-2v}{\sqrt{v}}.$$

Because the two integrands are priced by at least $\sigma - 1$ in (25),

$$\rho \le \rho_B(p) + D/(\sigma-1) < 1.1 + .047/(54/25) < 9/8.$$

Here $v \ge 15/64$ implies $\rho_B < 1.1$ by squaring positive quantities. The signed-second-moment identity gives

$$1 - 2p - 2q\theta \le \beta \le 1 - 2p + 2p\theta;$$

its endpoints imply $.066148 < \beta < .2527$.

To bound the scaled noise $\lambda$, retain the cubic cost rather than only its sign. If $Z$ is centered, $Z \le a$, $a > 0$, and $t = \mathbb{E}Z^2$, then

$$0 \le \mathbb{E}[(a - Z)(Z + t/a)^2] = at - \mathbb{E}Z^3 - t^2/a.$$

Apply this to $U_0$ and $-U_1$. A zero endpoint forces the corresponding noise to vanish, so no division by zero is needed. For $\lambda_0 = nq\mathbb{E}U_0^2$, $\lambda_1 = np\mathbb{E}U_1^2$, the same gap-budget expansion, before bounding these two cubics, reads

$$\begin{aligned} nD/\ell^3 \ge\ & nv\mathcal{Q} + 6nv\{d(1-d) + a_0a_1\} \\ & + \{3(\sigma+3)p - (\sigma+7)a_0\}\lambda_0 + \{3(\sigma+1)q - (\sigma+5)a_1\}\lambda_1 \\ & + \frac{(\sigma+1)\lambda_0^2}{nqa_0} + \frac{(\sigma-1)\lambda_1^2}{npa_1}. \end{aligned}$$

Terms with $a_j = 0$ are absent. Set $\bar{d} = .004$ and use $d(1-d) + a_0a_1 \ge (1-\bar{d})(a_0 + a_1)$. The reciprocal term is essential: pairing its variance-squared cost with the inward-displacement cost gives a linear cost in $\lambda$. Specifically, AM–GM gives

$$6nv(1-\bar{d})a_0 + \frac{(\sigma+1)\lambda_0^2}{nqa_0} \ge 2\lambda_0\sqrt{6(1-\bar{d})(\sigma+1)p}.$$

Thus

$$nD \ge \ell^3\{nv\mathcal{Q} + \lambda_0 Y_0(p) + \lambda_1 Y_1(p)\}, \tag{27}$$

where

$$\begin{aligned} Y_0 &= 3(\sigma+3)p - (\sigma+7)\bar{d} + 2\sqrt{6(1-\bar{d})(\sigma+1)p}, \\ Y_1 &= 3(\sigma+1)q - (\sigma+5)\bar{d} + 2\sqrt{6(1-\bar{d})(\sigma-1)q}. \end{aligned}$$

Each increases in its respective argument $p$ or $q$. At $p = 3/8$ and $q = 107/200$, lower bounds 3 and 2.6 for the roots give $Y_0 > 12.8$, $Y_1 > 11.8$. Since $\lambda_0 + \lambda_1 = \lambda$,

$$B_* \ge nD > 1.996^3(11.8)\lambda \quad (\lambda > 0), \qquad 1.996^3(11.8)(.35) > B_*.$$

This proves $\lambda < .35$; at zero noise it is immediate. □

**Proposition 6.2** (Two contact rays)**.** *Under Proposition 6.1, suppose $(P, H)$ is an attained fixed-target pair at $C_{\mathrm{E}}$ with nonnegative excess and true predecessor $C_{n-1} \le C_{\mathrm{E}}$. Then*

$$-.6 < U_0 \le a_0, \qquad -a_1 \le U_1 < .6.$$

*The support conclusion is uniform in the raw threshold $H$.*

*Proof.* Use the contact upper expression (53). The all-threshold Gaussian heat estimate and its radial derivative bound from the proof of Theorem D.1 give, for $t > \sqrt{3}$,

$$H_n(H, \pm t) \le h(t) := S^3\left\{\frac{2}{5}\left(\frac{t^3 - 3t}{6} + \frac{|t^4 - 6t^2 + 3|}{72}\right) + \frac{t^5}{4100} + \frac{3t}{820} + \frac{17(3t^2-2)}{78720}\right\},$$

$$|\partial_t H_n(H, \pm t)| < g(t) := \frac{2}{5}(t^2/2 + 1/2 + \eta), \qquad S = 81/80, \quad \eta = 8/875.$$

The first bound follows directly from (54): $\|\phi'''\|_\infty < 2\phi(0)$, $n-1 \geq 41$, and $\sqrt{n-1} > 6$. No derivative of the absolute-value majorant is taken.

At a support point of magnitude $t$, contact requires a nonnegative upper expression $H_n(H, \pm t) + C_{\mathrm{E}} Q_b(t)$, where

$$Q_b(t) = -t^3 + \tfrac{3}{2} r t^2 + 3bt + \eta r, \quad r = 9/8, \quad b = \begin{cases} -.066148 & \text{negative ray,} \\ .2527 & \text{positive ray.} \end{cases}$$

These replacements follow with their signs from (24); the discarded excess term is nonpositive. Take $c_0 = 819/2000 < C_{\mathrm{E}}$ and the two fixed boundaries

$$t_- = 1.996(3/8 + .6) = 1.9461, \qquad t_+ = 1.996(107/200 + .6) = 2.26546.$$

For a fixed raw threshold $H$, abbreviate the appropriate radial function $H_n(H, \pm t)$ by $H_n(t)$, with derivative taken in the positive magnitude $t$. Set

$$L(t) = g(t) + c_0 Q_b'(t).$$

Exact rational substitution gives the following conservative bounds:

| Boundary quantity | $t_-$, negative ray | $t_+$, positive ray |
|---|---|---|
| $h(t) + c_0 Q_b(t)$ | $< -2/5$ | $< -13/100$ |
| $L(t)$ | $< -1$ | $< -8/5$ |
| $L'(t)$ | $< -13/5$ | $< -16/5$ |

Both boundaries satisfy $t^2 > 3$. The constant second derivative is

$$L'' = \frac{2}{5} - 6c_0 = -\frac{2057}{1000} < 0.$$

Consequently $L'$ and then $L$ remain negative on each outward ray. Because $g > 0$, the identity $Q_b' = (L - g)/c_0$ gives $Q_b' < 0$ there. At the boundary $h \geq 0$, so the first table entry gives $Q_b < 0$; this sign also persists outward. Since $C_{\mathrm{E}} > c_0$, we obtain

$$H_n(t_\pm) + C_{\mathrm{E}} Q_b(t_\pm) < 0, \qquad H_n'(t) + C_{\mathrm{E}} Q_b'(t) \leq g(t) + c_0 Q_b'(t) = L(t) < 0.$$

The contact upper expression therefore cannot be nonnegative anywhere on either ray. This excludes every support point there, including boundary atoms. Only the actual radial function is differentiated; the absolute-value majorant $h$ is used solely at the boundary.

On the negative class, $X > -t_-$ and $\ell > 1.996$ give $U_0 = X/\ell + p > -(3/8 + .6) + p > -.6$. On the positive class, $X < t_+$ gives $U_1 = X/\ell + p - 1 < 107/200 + .6 + p - 1 < .6$. The inward endpoints and conditional centering are the actual ones from Proposition 6.1. □

The pointwise normalization is worth recording once:

$$1 = v\ell^2 + \theta, \qquad s = \mathbb{E}U^2 = \theta/\ell^2, \qquad v + s = \ell^{-2}, \qquad X = \frac{B_0 + U - p}{\sqrt{v+s}}. \tag{28}$$

Thus the law to be compared in the next section is exactly the original law, and its threshold remains the raw $H$. Reflection, when required, selects the oriented positive witness before acquisition; neither its signed defect budget nor its probability interval is transported from the opposite orientation.

# 7 The direct comparison with a binomial law

In count coordinates, conditioning on the number of positive summands expresses the sum as an integer count plus centered noise. We compare its CDF with the corresponding binomial CDF. Noise can move mass out of the count just below the threshold and bring mass in from counts above it. The proof shows that the outgoing loss, together with the normal and third-moment corrections, exceeds all incoming contributions. The quantity $\lambda = n\mathbb{E}U^2$ measures the total noise variance in count units.

**Theorem 7.1** (The common-strip comparison)**.** *Let $n \geq 700$ be an integer, $B \sim \mathrm{Bernoulli}(p)$, and $\mathbb{E}(U \mid B) = 0$, where $p \in [3/8, 93/200]$. Write $U_j$ for the conditional noise given $B = j$, and suppose*

$$-3/5 \leq U_0 \leq a_0, \qquad -a_1 \leq U_1 \leq 3/5, \qquad a_0, a_1 \geq 0, \quad a_0 + a_1 \leq 1/250.$$

*Put $q = 1 - p$, $v = pq$, $s = \mathbb{E}U^2$, $\lambda = ns$, $X = (B + U - p)/\sqrt{v + s}$ and $\rho = \mathbb{E}|X|^3$. For independent copies and every raw threshold $|H| \leq 46/25$,*

$$\sqrt{n}\left\{\mathbb{P}\left(\sum_{i=1}^{n} X_i \leq H\right) - \Phi(H/\sqrt{n})\right\} - C_{\mathrm{E}}\rho < -\lambda/100 \qquad (0 < \lambda \leq 7/20). \tag{29}$$

*At $\lambda = 0$ the same excess is strictly negative.*

The proof below uses only the displayed hypotheses. Contact and the sharp predecessor are needed earlier to acquire this strip, not for its local comparison. The small- and larger-noise intervals meet at $\lambda = 1/100$. For small noise, we keep every error proportional to $\lambda$ and use the loss from one count. For larger noise, five neighboring counts give a uniform crossing loss. That second estimate has an error of order $1/n$ which does not vanish at $\lambda = 0$, so it cannot cover arbitrarily small positive noise. At zero noise the law is exactly Bernoulli.

## 7.1 Three elementary ingredients

Write $\pi_j = \binom{n}{j} p^j q^{n-j}$, $F_B(j) = \sum_{i \leq j} \pi_i$, and $\rho_B = (1 - 2v)/\sqrt{v}$. We use Schulz's strict finite-binomial theorem [13, Theorem 1] in the form

$$\sqrt{n}\{F_B(j) - \Phi((j - np)/\sqrt{nv})\} < C_{\mathrm{E}}\rho_B \quad (0 < p < 1, n < \infty). \tag{30}$$

Its strictness concerns every finite binomial law; it is not inferred from the limiting lower approximation for the universal constant.

The factorial estimates used throughout are finite Stirling bounds:

$$\log(j!) = (j + \tfrac{1}{2})\log j - j + \tfrac{1}{2}\log(2\pi) + r_j, \qquad 0 < r_j < 1/(12j).$$

For completeness, with $x = 1/(2j + 1)$, $r_j - r_{j+1} = \operatorname{arctanh}(x)/x - 1 = \sum_{l \geq 1} x^{2l}/(2l + 1)$ lies strictly between 0 and $1/(12j(j + 1))$. Wallis's product fixes the limiting constant, so $r_j \to 0$; summing these differences proves the bounds. One useful consequence is

$$\max_j \mathbb{P}\{\mathrm{Bin}(M, p) = j\} < \frac{41}{100\sqrt{Mv}} \quad (M \geq 43, p \in [1/3, 2/3]). \tag{31}$$

Indeed a mode has $z = j/M = p + \delta/M$, $|\delta| \leq 2/3$. Since $v \geq 2/9$ and $|1 - 2p| \leq 1/3$, $z(1 - z) \geq v(1 - 1/M - 2/M^2) > 39v/40$. Stirling, the nonnegative relative entropy, and $e^{1/516} < 516/515$ bound the mass by $(2/5)\sqrt{40/39}(516/515)/\sqrt{Mv}$, which is smaller than the right side of (31) by a rational square comparison.

Finally the cubic correction has the favorable sign on the full strip. Let $s_j = \mathbb{E}U_j^2$, so $s = qs_0 + ps_1$, and set $r = 3/8$, $d = 1/250$, $v_* = 15/64$. The conditional supports have fixed signs relative to $p$, since $d < r$. Expansion of the raw third absolute moment gives

$$\mathbb{E}|B + U - p|^3 = v(1-2v) + 3v(s_0 + s_1) - q\mathbb{E}U_0^3 + p\mathbb{E}U_1^3.$$

The inward bounds imply $\mathbb{E}U_0^3 \le a_0 s_0$, $\mathbb{E}U_1^3 \ge -a_1 s_1$. Together with $(1+x)^{3/2} \le 1 + 3x/2 + 3x^2/8$ for $x \ge 0$, this yields

$$\begin{aligned}(v+s)^{3/2}(\rho - \rho_B) &\ge s\left\{3r - d - \frac{3}{2}(1-2v) - \frac{3(1-2v)s}{8v}\right\} \\ &\ge s\left\{\frac{2593}{8000} - \frac{17}{20}s\right\} > 0, \qquad 0 < s \le \frac{1}{2000}. \end{aligned} \tag{32}$$

Here $s \le (7/20)/700 = 1/2000$, and the displayed coefficient is $3r - d - \frac{3}{2}(1-2v_*)$. Thus every nonnegative coefficient of $\rho_B - \rho$ can be discarded in an upper comparison.

### 7.2 Small noise: one count and the whole incoming tail

Assume $0 < \lambda \le L = 1/100$. Choose an integer $k$ with

$$t = np + H\sqrt{v+s} = k + u, \qquad -3/10 \le u \le 7/10, \qquad \delta = k - np.$$

The phase interval has length one, so such an integer always exists. It need not be the floor anchor used in the larger-noise proof. Let $W_j$ have the law of the sum of $j$ independent $U_1$ variables and $n-j$ independent $U_0$ variables. For $C \ge 0$, subtract the signed binomial excess at $k$ from the actual excess:

$$\begin{aligned}\Delta_C = \sqrt{n}\left\{\mathbb{P}\left(\sum X_i \le H\right) - \Phi(H/\sqrt{n})\right\} - C\rho \\ - \sqrt{n}\{F_B(k) - \Phi(\delta/\sqrt{nv})\} + C\rho_B.\end{aligned}$$

Exact conditioning gives

$$\begin{aligned}\mathbb{P}\left(\sum X_i \le H\right) - F_B(k) = &-\sum_{i=0}^{k} \pi_{k-i}\mathbb{P}(W_{k-i} > u + i) \\ &+ \sum_{i=1}^{n-k} \pi_{k+i}\mathbb{P}(W_{k+i} \le u - i).\end{aligned}$$

Outgoing events are strict and incoming events closed; this convention includes all threshold atoms. Define

$$f(u) = \frac{\Phi((\delta+u)/\sqrt{n(v+s)}) - \Phi(\delta/\sqrt{n(v+s)})}{\pi_k}, \qquad O = \mathbb{P}(W_k > u) + f(u),$$

$$T_{\text{in}} = \sum_{i=1}^{n-k} \mathbb{P}(W_{k+i} \le u - i), \qquad \mathcal{R} = \sqrt{n}\{\Phi(\delta/\sqrt{nv}) - \Phi(\delta/\sqrt{n(v+s)})\} + C(\rho_B - \rho).$$

Keeping the own-count loss and discarding the other nonpositive outgoing terms gives

$$\Delta_C \le -m_k(O - RT_{\text{in}}) + \mathcal{R}, \qquad m_k = \sqrt{n}\pi_k, \tag{33}$$

whenever $\pi_{k+i}/\pi_k \le R$ for all legal incoming counts.

**Uniform normalization.**

Set $D_* = 5/3$, $b_* = 1/20$, $T_* = 101/10000$ and $c_* = 2/525$. Because $v \geq v_*$ and $v + s < 1/4$, $|t - np| < 23/25$. Consequently $|\delta| < D_*$ globally and $|\delta| < 1$ when $|u| \leq b_*$. Write $\mathcal{K}(z, p) = z \log(z/p) + (1-z)\log((1-z)/q)$ for binomial relative entropy; this is distinct from the physical gap defect $D$. The segment between $p$ and $k/n$ lies inside $[.37, .47]$; in particular $1 \leq k \leq n-1$. On this segment

$$z(1-z) > 23/100, \quad |\partial_z^3 \mathcal{K}(z,p)| < 5, \quad |(\log(z(1-z)))'| < 6/5, \quad (12z(1-z))^{-1} < 3/8.$$

Stirling and entropy Taylor expansion therefore give

$$m_k = \frac{\phi(0)}{\sqrt{v}} e^{-\delta^2/(2nv)+e}, \qquad n|e| < Z(D_0) := 3D_0/5 + 3/8 + D_0^3/840 \tag{34}$$

for $|\delta| \leq D_0 \leq D_*$. The three terms respectively bound the logarithmic prefactor, factorial remainder and entropy remainder. Also

$$\log f'(z) = -\tfrac{1}{2}\log(1 + s/v) + \frac{\delta^2}{2nv} - \frac{(\delta+z)^2}{2n(v+s)} - e,$$

so, on an integration interval $|z| \leq b_0$,

$$n|\log f'(z)| \leq \frac{11}{5}\lambda + \frac{2D_0 b_0 + b_0^2}{2v_*} + Z(D_0).$$

The coefficient $11/5$ exceeds $1/(2v_*) + D_*^2/(1400v_*^2)$. Using $e^{-x} \geq 1 - x$, $e^x \leq (1-x)^{-1}$ and $\phi(0) > 199/500$ now gives

| | $\lvert u\rvert \leq 1/20$ | $-3/10 \leq u \leq 7/10$ |
|---|---|---|
| $m_k$ lower bound | 3/4 | 3/4 |
| $f'$ lower bound | 99/100 | 49/50 |
| $f'$ upper bound | 101/100 | 51/50 |
| $R$ upper bound | 51/50 | 6/5 |

(35)

The derivative estimates hold on the whole segment from 0 to $u$. For the global ratio, (31) gives $\sqrt{n}\max_j \pi_j < 17/20$, hence $R < (17/20)/(3/4) < 6/5$. In the inner interval the first ratio is

$$\frac{\pi_{k+1}}{\pi_k} = 1 + \frac{p - \delta - 1}{nv + q(\delta+1)} < 1 + \frac{93/200}{700 v_*} < 51/50;$$

every later ratio is at most one, since $p - \delta < 2$. By (32) the moment correction is nonpositive. Differentiating the normal CDF and using $\phi(0)/(2v_*^{3/2}) < 2$ therefore gives

$$\mathcal{R} < 2\lambda/n \quad (|u| \leq b_*), \qquad \mathcal{R} < 4\lambda/n \quad \text{on the full phase interval.} \tag{36}$$

Indeed the positive displacement is bounded by 1 or $D_* < 2$.

**No truncation of the incoming tail.**

Since $|s_1 - s_0| \leq (8/3)s$, the actual own-count variance $t_0 = \mathbb{E}W_k^2$ satisfies $.99\lambda < t_0 < 1.01\lambda \leq T_*$. For each incoming count,

$$t_i = \mathbb{E}W_{k+i}^2 \leq \lambda\{1 + c_*(D_0 + i)\}, \qquad \mathbb{E}W_{k+i}^4 \leq A^2 t_i + 3t_i^2, \quad A = 3/5,$$

where $D_0 = 1$ in the inner interval and $D_0 = D_*$ otherwise. The moment bound follows from independence and $|U_j| \le A$. For $a = 1 + c_* D_0$ put

$$K(a) = A^2 \left( \frac{13a}{12} + \frac{5c_*}{4} \right) + 3L \left( \frac{13a^2}{12} + \frac{5ac_*}{2} + \frac{5c_*^2}{3} \right).$$

Fourth-moment Markov at distance $i - u$, and extension of the nonnegative sum to all positive integers, give

$$\begin{aligned} T_{\text{in}} &< \frac{K(1+c_*)}{(1-b_*)^4}\lambda < .53\lambda \quad && (|u| \le b_*), \\ T_{\text{in}} &< \frac{.43\lambda}{(1-u)^4} && (0 \le u \le .7), \\ T_{\text{in}} &< .43\lambda && (u \le 0). \end{aligned} \tag{37}$$

Here $K(1 + c_* D_*) < .43$, and $\sum i^{-4} < 13/12$, $\sum i^{-3} < 5/4$, $\sum i^{-2} < 5/3$. These series bounds follow by retaining respectively five, three and five terms and estimating the remaining sum by an integral. For $u \ge 0$ use $i - u \ge (1-u)i$; for $u \le 0$ use $i - u \ge i$.

**The own-count loss at the central phases.**

Set $\alpha = 1/2$ and $m_h = 5\alpha^4 = 5/16$. The factored quartic

$$h_\alpha(x) = \frac{-16x^4 + 12x^2 + 4x}{5} = \frac{4}{5} x(1-x)(2x+1)^2 \le \mathbf{1}_{(0,\infty)}(x) \tag{38}$$

is a global minorant for the strict event. Its factorization makes it nonpositive for $x \le 0$ and $x \ge 1$. On $(0,1)$ the maximum occurs at $x = (1+\sqrt{3})/4$ and equals $(9 + 6\sqrt{3})/20 < 1$; the last comparison follows from $6\sqrt{3} < 11$. Thus it may be integrated against the conditional sum $W_k$, which need not be bounded by $3/5$. Only its individual summands have that bound. The expectation of the quartic uses moments through order four, supplied by independence and the bounded summands. Let $r = zt_0$ and set

$$\begin{aligned} F_{\varepsilon,b}(z,t) = \frac{3\alpha^2 - A^2 - 3t}{m_h} + \left( \frac{2\varepsilon}{5\alpha} + b - \frac{4At}{m_h} \right) z \\ + \frac{3\alpha^2 t - 6t^2}{m_h} z^2 - \frac{t^3 z^4}{m_h}. \end{aligned}$$

Expanding $h_\alpha(W_k + r)$ for $u = -r$, or $h_\alpha(W_k - r)$ for $u = r$, and using $|\mathbb{E}W_k^3| \le At_0$, $\mathbb{E}W_k^4 \le A^2 t_0 + 3t_0^2$, yields

$$O/t_0 \ge F_{1,-101/100}(z, t_0) \quad (u = -r), \qquad O/t_0 \ge F_{-1,99/100}(z, t_0) \quad (u = r).$$

Both functions exceed $3/5$ on $0 \le z \le 3/2$, $0 \le t \le T_*$. Indeed $\partial_t^2 F = -(12z^2 + 6tz^4)/m_h \le 0$, so it suffices to bound the two edges $t = 0, T_*$. Replacing $z^4$ by $(9/4)z^2$ in its negative term gives a lower quadratic whose coefficient of $z^2$ is

$$q_{\text{quad}}(t) = \frac{3\alpha^2 t - 6t^2 - (9/4)t^3}{m_h}, \qquad q_{\text{quad}}(0) = 0, \quad q_{\text{quad}}(T_*) = \frac{27842487291}{1250000000000} > 0.$$

For the negative phase, its derivative at the right endpoint is $-21/100$ when $t = 0$, and is less than $-1/5$ when $t = T_*$. The quadratic therefore decreases throughout the interval. For the positive phase, its derivative at zero is $19/100$ or $7027/62500$, so it increases. The resulting endpoint minima satisfy

| | $t = 0$ | $t = T_*$ |
|---|---|---|
| $F_{1,-101/100}$ lower bound at $z = 3/2$ | $933/1000$ | $> 769/1000$ |
| $F_{-1,99/100}$ lower bound at $z = 0$ | $156/125$ | $3597/3125$ |

All four exceed 3/5. Concavity in $t$ then covers the entire rectangle.

For negative phases $3t_0/2 \le r \le b_*$, Cantelli gives

$$O \ge \frac{r^2}{t_0 + r^2} - \frac{101}{100} r > \frac{3}{5} t_0.$$

Its derivative at $r = 3t_0/2$ is positive and first increases, then decreases. The minimum is therefore at an endpoint. At the first use $t_0 \le T_*$ after division by $t_0$; at the other use $b_*^2/(T_* + b_*^2) - (101/100)b_* > (3/5)T_*$. For positive phases beyond this switch, $O \ge (99/100)r \ge (297/200)t_0 > (3/5)t_0$. The switch lies below $b_*$ because $3T_*/2 < b_*$. Thus

$$O > \frac{3}{5} t_0 > .59\lambda \qquad (|u| \le b_*). \tag{39}$$

The value $u = 0$ is included, with $h_\alpha(0) = 0$.

**The outer phases and the final margin.**

For $u = -r$, $b_* \le r \le 3/10$, Cantelli gives

$$O \ge \frac{r^2}{T_* + r^2} - \frac{51}{50} r > 7/50 \ge 14\lambda.$$

The derivative is positive at $b_*$ and has at most one downward zero; both endpoint values exceed 7/50. For $b_* \le u \le 7/10$,

$$O - RT_{\text{in}} \ge \frac{49}{50} u - \frac{(6/5)(43/100)L}{(1-u)^4} > 1/25 \ge 4\lambda.$$

This function is concave and both endpoints exceed 1/25. In the inner interval, (37) and (39) give $O - RT_{\text{in}} > (.59 - (51/50).53)\lambda$. Substitute these three margins and (36) into (33), using $m_k > 3/4$. The smallest final margin is the inner one, and

$$\frac{3}{4}\left(\frac{59}{100} - \frac{51}{50}\frac{53}{100}\right) - \frac{2}{700} > \frac{1}{100}.$$

Hence $\Delta_C < -\lambda/100$. Taking $C = C_{\text{E}}$ and applying (30) proves (29) for $0 < \lambda \le 1/100$.

### 7.3 Larger noise: five counts and two endpoint checks

Now assume $1/100 \le \lambda \le 7/20$ and choose $k = \lfloor t \rfloor$, $u = t - k \in [0, 1)$, $w = t - np$. The following constants will be used only in this subsection:

$$N = 700, \quad p_- = 3/8, \quad p_+ = 93/200,$$
$$r = 3/8, \quad q_{\min} = 107/200,$$
$$v_- = 15/64, \quad v_+ = 9951/40000, \quad A = 3/5, \quad d = 1/250,$$
$$\eta = 2/5, \quad c_{\text{mass}} = 41/100, \quad \pi_+ = 22/7, \quad \phi_- = 199/500, \quad \phi_+ = 2/5,$$
$$\kappa = 3, \quad \xi = 93/100, \quad c_- = 6, \quad \zeta = 23/100, \quad m_* = 3/4, \quad V_* = 113/1000,$$
$$\bar{c} = 79/200, \quad j_0 = 1/5, \quad j_1 = 5/2.$$

In particular $|w| < \xi < 1$, since $(46/25)^2(v_+ + (7/20)/N) < \xi^2$. Also $N(r - d) > \xi$, so $na_0 \le t < n(1 - a_1)$.

**An exact weighted crossing identity.**

On an auxiliary space take independent copies of the conditional laws and write

$$T_j = j + \sum_{l \le j} U_{1l} + \sum_{l > j} U_{0l}, \qquad L_j(t) = \mathbb{E}(t - T_j)_+, \qquad V_i = \sum_{l < i} U_{1l} + \sum_{l > i} U_{0l}.$$

The mesh is increasing because $T_j - T_{j-1} \ge 1 - a_0 - a_1 > 0$. Each $T_j$ has the conditional convolution given $j$ successes. Set

$$h(x) = \mathbb{E}(x - U_0)_+ - \mathbb{E}(x - 1 - U_1)_+ - \mathbb{P}(1 + U_1 \le x), \qquad g_i = \mathbb{E}h(t - i + 1 - V_i).$$

**Lemma 7.2** (Count interpolation and crossing loss)**.** *With the preceding definitions,* $B_j = L_j(t) - (t - j)_+ \ge 0$ *and*

$$\mathbb{P}\left(\sum X_i \le H\right) = (1 - u)F_B(k) + uF_B(k + 1) + \sum_{j=1}^{n-1} (\pi_{j+1} - \pi_j)B_j - \sum_{i=1}^{n} \pi_i g_i. \tag{40}$$

*Proof.* Jensen's inequality for the convex positive-part function and $\mathbb{E}T_j = j$ give $B_j \ge 0$. Independence of the omitted coordinate gives $g_i = L_{i-1}(t) - L_i(t) - \mathbb{P}(T_i \le t)$. Here $L_0 = t$, $L_n = 0$ and $\mathbb{P}(T_0 \le t) = 1$: the support bounds give $T_0 \le na_0 \le t < T_n$. Multiplication by $\pi_i$ and finite summation by parts now gives the identity, since replacing $L_j$ by $(t - j)_+$ in $\pi_0 + \pi_1 t + \sum(\pi_{j+1} - \pi_j)L_j$ telescopes to the stated interpolation. All CDF events are closed, including atoms. □

The three terms in (40) are the interpolated binomial probability, an error from variation in the binomial masses, and the crossing loss. The next argument proves $g_i \ge 0$ using the separated conditional supports and conditional centering; this does not follow merely from $B_j \ge 0$.

**Five kernels with four centered replacements.**

The sine bump below serves three purposes. Its translates recover a single periodic function, its Lipschitz derivative makes centered replacement errors proportional to variance, and the identity for $\sin^2$ turns independence into a product of characteristic functions. Five translates will supply the main loss; the omitted translates are bounded using a fourth moment. Let $f(x) = \sin^2(\pi x)$ on $[0, 1]$ and zero outside. It is continuously differentiable with derivative Lipschitz constant $2\pi^2$. We claim

$$h(x) \ge \eta f(x) \ge 0. \tag{41}$$

For $x < 1 - a_1$, $h(x) = \mathbb{E}(x - U_0)_+ \ge 0$, and Jensen gives $h(x) \ge x$ when $x \ge 0$. On $[0, 1]$, $f(x) \le (5/2)x$: use $\pi^2 < 10$ up to $1/4$, the concave tangent $1/2 + \pi(x - 1/4)$ on $[1/4, 2/5]$, and $f \le 1$ thereafter. For $x \ge 1 - a_1 > a_0$, put $D_1 = 1 + U_1$ and $L_1 = \mathbb{E}(D_1 - x)_+$. Put–call parity gives $h(x) = \mathbb{P}(D_1 > x) - L_1 \ge 0$, since $D_1 \le 1 + A$ and $x > A$. When $1 - a_1 \le x \le 1$, $L_1 \ge 1 - x$ and

$$h(x) \ge \frac{x - A}{1 + A - x} L_1 \ge \frac{1 - A - d}{A + d}(1 - x) \ge \eta\pi^2(1 - x)^2 \ge \eta f(x).$$

The penultimate inequality uses $\eta\pi_+^2 d \le (1 - A - d)/(A + d)$. At $x = 0, 1$ both comparisons are weak, as required for atoms.

Freeze $V = V_{k+1}$ and let $V^{(j)}$ have the law of $V_{k+1+j}$. For $j > 0$ it replaces $j$ copies of $U_0$ by $U_1$. With $f_j(x) = f(u - j - x)$, regroup the two positive changes as

$$\begin{aligned}&\mathbb{E}f_1(V^{(1)}) + \mathbb{E}f_2(V^{(2)}) - \mathbb{E}f_1(V) - \mathbb{E}f_2(V)\\ &\quad = \mathbb{E}(f_1 + f_2)(V^{(1)}) - \mathbb{E}(f_1 + f_2)(V) + \mathbb{E}f_2(V^{(2)}) - \mathbb{E}f_2(V^{(1)}).\end{aligned}$$

There is an identical grouping for the negative changes. Adjacent bumps have disjoint open supports and zero derivatives at the joins, so the derivative Lipschitz constant remains $2\pi^2$. For a centered $Z$, $|\mathbb{E}F(a + Z) - F(a)| \le \frac{1}{2}\operatorname{Lip}(F')\mathbb{E}Z^2$. Conditioning on all unchanged summands, and applying this twice to each replacement, bounds the total cost by $4\pi^2(s_0 + s_1)$. Hence

$$\sum_{i=k-1}^{k+3} g_i \ge \eta\left\{\sum_{j=-2}^{2} \mathbb{E}f(u - j - V) - 4\pi^2(s_0 + s_1)\right\}.$$

These five indices are interior, because $|i - np| < \xi + 3 < 4$. No independence between different count sums is asserted.

The infinite translates of $f$ sum to $\sin^2(\pi(u - V))$. An omitted nonzero translate requires $|V| > 2$, including endpoint collisions because the bump vanishes there. Thus the omitted part is at most $\mathbb{E}V^4/16$. With $\delta = k - np \in (-1 - \xi, \xi)$, the frozen variance is

$$\tau = (nq - \delta - 1)s_0 + (np + \delta)s_1, \qquad (1 - 6/n)\lambda \le \tau \le (1 + 3/n)\lambda.$$

The lower estimate uses $c_- r \ge 1 + \xi$; the upper uses $r \ge 1/3$. Also $s_0 + s_1 \le \lambda/(nr)$ and $\mathbb{E}V^4 \le A^2\tau + 3\tau^2$. For either conditional noise $Z$, an independent copy satisfies $|Z - Z'| \le A + d < 2/3$. Concavity of sine and $\sin(2\pi/3) = \sqrt{3}/2 > 17/20$ give $\sin(\pi x) \ge (51/40)x$ on $[0, A + d]$; hence $1 - \cos(2\pi x) \ge 3x^2$ on $|x| \le A + d$. Therefore

$$1 - |\mathbb{E}e^{2\pi i Z}|^2 = \mathbb{E}\{1 - \cos(2\pi(Z - Z'))\} \ge 2\kappa \operatorname{Var} Z,$$

and $|\mathbb{E}e^{2\pi i Z}| \le e^{-\kappa \operatorname{Var} Z}$. Multiplying over the independent frozen summands shows $\mathbb{E}\sin^2(\pi(u - V)) \ge (1 - e^{-\kappa\tau})/2$. Altogether,

$$\sum_{i=k-1}^{k+3} g_i \ge \eta\left\{\frac{1 - e^{-\kappa(1-6/n)\lambda}}{2} - \frac{A^2(1 + 3/n)\lambda + 3(1 + 3/n)^2\lambda^2}{16} - \frac{4\pi^2\lambda}{nr}\right\}. \tag{42}$$

**The weights and the complete curvature tail.**

For $k - 1 \le i \le k + 3$, the segment between $i/n$ and $p$ has variance product greater than $\zeta$; indeed $\zeta < \min_{z=p_- - 4/N,\, p_+ + 4/N} z(1 - z)$. Relative entropy is bounded by $(i - np)^2/(2n\zeta) < 8/(n\zeta)$, and the factorial remainder is greater than $-1/(12n\zeta)$. Stirling therefore gives

$$\sqrt{n}\pi_i > 2\phi_-\left(1 - \frac{8 + 1/12}{N\zeta}\right) > m_*.$$

Every other $g_i$ is nonnegative. The reserve in (42) increases with $n$. Replace $n$ by $N$ first, then use $1 - e^{-x} \ge x - x^2/2 + x^3/6 - x^4/24$ for $x \ge 0$. Writing $\alpha = 1 - 6/N$, $b = 1 + 3/N$ and $z = \kappa\alpha$, we obtain

$$\sqrt{n}\sum_i \pi_i g_i \ge W(\lambda) := \eta m_*\left\{\frac{z\lambda - z^2\lambda^2/2 + z^3\lambda^3/6 - z^4\lambda^4/24}{2} - \frac{A^2 b\lambda + 3b^2\lambda^2}{16} - \frac{4\pi_+^2\lambda}{Nr}\right\}. \tag{43}$$

Put $c = (np - q - t)_+$. For $a = t - j$ and $W_j = T_j - j$, centering gives

$$B_j = (\mathbb{E}|a - W_j| - |a|)/2 \ge 0, \qquad B_j \le \sqrt{T}/2, \qquad a_+ B_j \le T/4,$$

whenever $T \ge \mathbb{E}W_j^2$. The factorization $27x^4 - 256x + 256 = (3x-4)^2(3x^2+8x+16)$ gives $B_j \le 27M_4/(256|a|^3)$ for $a \ne 0$, if $M_4 \ge \mathbb{E}W_j^4$. The exact adjacent-mass formula is

$$\pi_{j+1} - \pi_j = \frac{np - q - j}{(n+1)v}\,\mathbb{P}\{\mathrm{Bin}(n+1,p) = j+1\}.$$

Discard its nonpositive terms and bound the positive numerator by $(a+c)_+$. Since $c \le \bar{c} < 1$, at most one mesh point lies in each of $(-c,0)$ and $[0,1)$; together they cost $c\sqrt{T} + T/4$. The rest is bounded by the complete series $\sum i^{-2} < 5/3$, $\sum i^{-3} < 5/4$. Using (31) and $\sqrt{n}/(n+1)^{3/2} \le 1/n$ yields

$$\sqrt{n}\sum_j (\pi_{j+1} - \pi_j)B_j \le \frac{c_{\mathrm{mass}}}{nv^{3/2}}\left\{\frac{T}{4} + c\sqrt{T} + \frac{27M_4}{256}\left(\frac{5}{3} + \frac{5c}{4}\right)\right\}. \tag{44}$$

A positive adjacent mass requires $j < np$, where its actual variance is at most $\lambda/q \le T = \lambda/q_{\min}$ and $\mathbb{E}W_j^4 \le M_4 = A^2T + 3T^2$. Moreover $L_{\mathrm{var}}(\lambda) = j_1\lambda + j_0 \ge \sqrt{T}$, since $4j_0j_1 \ge 1/q_{\min}$.

**Both signs of the normal correction.**

Apply (30) at $k, k+1$ and interpolate. For $G(x) = \Phi((x-np)/\sqrt{nv})$, $G''(x) \le \phi(0)(np - x)_+/(nv)^{3/2}$. The interpolation remainder is

$$\sqrt{n}\{(1-u)G(k) + uG(k+1) - G(t)\} \le \frac{\phi(0)}{2nv^{3/2}}u(1-u)(u-w)_+.$$

If $H \le 0$, changing normal variance from $v$ to $v+s$ is favorable. Using $u(1-u) \le 1/4$ and $u^2(1-u) \le 4/27$, the total normal cost is at most $\phi_+(\xi/8 + 2/27)/(nv^{3/2})$, and $c = (-q-w)_+ \le \bar{c}$. If $H \ge 0$, $c = 0$, interpolation costs at most $\phi_+(2/27)/(nv^{3/2})$, and the variance change costs at most $\phi(0)ws/(2v^{3/2})$ by $1 - (1+x)^{-1/2} \le x/2$. The cubic correction is favorable by (32).

Define the nonnegative quadratics

$$S_c = \frac{T}{4} + cL_{\mathrm{var}}(\lambda) + \frac{27M_4}{256}\left(\frac{5}{3} + \frac{5c}{4}\right), \qquad E_- = \frac{c_{\mathrm{mass}}S_{\bar{c}} + \phi_+(\xi/8 + 2/27)}{V_*},$$

$$E_+ = \frac{c_{\mathrm{mass}}S_0 + \phi_+(2/27 + \xi\lambda/2)}{V_*}.$$

Because $V_*^2 < v_-^3$, combining (40)–(44) with these normal bounds makes the full excess strictly less than $-W(\lambda) + E_\pm(\lambda)/n$, with the sign chosen according to $H$. A single envelope suffices for both signs, because

$$\begin{aligned} V_*(E_- - E_+) &= c_{\mathrm{mass}}\bar{c}L_{\mathrm{var}}(\lambda) + c_{\mathrm{mass}}\frac{27M_4}{256}\frac{5\bar{c}}{4} + \phi_+\xi\left(\frac{1}{8} - \frac{\lambda}{2}\right) \\ &= \frac{7889}{100000} + \frac{1751}{8000}\lambda + c_{\mathrm{mass}}\frac{27M_4}{256}\frac{5\bar{c}}{4} > 0 \qquad (\lambda \ge 0). \end{aligned}$$

Set $E = E_-$. The full excess is strictly less than $-W(\lambda) + E(\lambda)/n$ for either sign of $H$.

**Concavity reduces the interval to two endpoints.**

Set $P(\lambda) = W(\lambda) - E(\lambda)/N - \lambda/100$. The nonnegative polynomial $E$ is a convex quadratic, whereas direct differentiation gives

$$W''(\lambda) = -\eta m_*\left\{\frac{z^2}{4}\left[(z\lambda - 1)^2 + 1\right] + \frac{3b^2}{8}\right\} < 0.$$

Thus $P$ is strictly concave. Substitution of the rational constants above gives the exact rational lower bounds

$$P(1/100) > .002314, \qquad P(7/20) > .061304.$$

Both are positive, proving $P > 0$ on the whole closed interval. Since $n \geq N$ and $E \geq 0$, this proves (29) for $1/100 \leq \lambda \leq 7/20$. The scalar checks used here involve only rational inequalities and $199/500 < \phi(0) < 2/5$, $25/8 < \pi < 22/7$; no parameter grid or tail truncation is involved.

The two positive-noise proofs have identical support and probability hypotheses, include their shared endpoint, and cover every phase. At $\lambda = 0$, conditional centering and $\mathbb{E}U^2 = 0$ give $U = 0$ almost surely, so (30) applies directly. This completes the proof of Theorem 7.1.

# 8 Composition and the finite boundary

**Theorem 8.1** (Acquired-witness exclusion)**.** *Let $n \geq 700$ and let $(P, H)$ attain the fixed-target maximum at $C_{\mathrm{E}}$ with $a(P, H) \geq 0$. Suppose $C_{n-1} \leq C_{\mathrm{E}}$, the structural inequality* (51)*, and the complete entry Conditions D.3 and D.5 hold. Then such a pair cannot exist.*

*Proof.* Theorem 5.1 gives $nD \leq 32.20665378$ and $|H| \leq 1.83605680$ for this same oriented, compactly supported law. Propositions 6.1 and 6.2 give the central gap and the following bounds. In (4), identify the local theorem's $B$ with $B_0 = \mathbf{1}_{\{X>0\}}$. The variable $(B + U - p)/\sqrt{pq + s}$ is exactly the original $X$, so the threshold remains the original raw $H$. The hypotheses match as follows.

| Bound just obtained | Hypothesis of Theorem 7.1 |
|---|---|
| $3/8 < p < 93/200$ | $p \in [3/8, 93/200]$ |
| $a_0, a_1 \geq 0$, $a_0 + a_1 < 1/250$ | Nonnegative inward endpoints, sum at most $1/250$ |
| $-3/5 < U_0 \leq a_0$, $-a_1 \leq U_1 < 3/5$ | The stated closed conditional-support bounds |
| $0 \leq \lambda < 7/20$ | Positive-noise interval or separate zero-noise case |
| $\lvert H\rvert \leq 1.83605680 < 46/25$ | The full threshold interval |
| Independent copies of $X$ | The iid sum in the local comparison |

Positive noise gives $a(P, H) < -\lambda/100$; zero noise is Bernoulli and gives strictly negative excess by Schulz. Both contradict $a \geq 0$. □

**Theorem 8.2** (Conditional sharp inequality)**.** *Assume a genuine finite base $C_j \leq C_{\mathrm{E}}$ for $1 \leq j \leq 699$, the external structural and strict finite-binomial theorems used above, and the complete entry predicates at every least failure at $n \geq 700$. Then*

$$C_n < C_{\mathrm{E}} \ (n \geq 700), \qquad C_n \leq C_{\mathrm{E}} \ (n \geq 1), \qquad \sup_{n \geq 1} C_n = C_{\mathrm{E}}.$$

*Proof.* If $C_n < C_{\mathrm{E}}$ fails above the base, choose the least failing index. Proposition 2.1 and the reflection argument in Section 2.2 supply an attained positive closed-event pair with $a \geq 0$. The finite base and minimality give the true predecessor $C_{n-1} \leq C_{\mathrm{E}}$. The preceding theorem excludes the pair. The independent Bernoulli limit (2) gives equality for the global supremum. □

With a base only through 41, the same argument locates any first failure in $42, \dots, 699$. An unresolved earlier failure prevents the predecessor argument at later indices. Reflection also includes the exact atom rule:

$$\mathbb{P}(-S_n \le -H) - \Phi(-H/\sqrt{n}) = \Phi(H/\sqrt{n}) - \mathbb{P}(S_n < H) \ge \Phi(H/\sqrt{n}) - \mathbb{P}(S_n \le H).$$

The difference is the atom at $H$. The selected orientation acquires its own signed $D$ budget; that budget is not copied under reflection.

**Role of finite verification.**

The entry conditions retain both coarse passes, all four signed passes, and every incidence in the internal 100-panel contract. The local comparison uses the resulting common support region. Its scalar checks are separate from entry verification and from the finite-base assumption; Appendix E gives their commands and scope.

# 9 Discussion

Fixing the target aligns the coefficient in the extremal problem, contact, and the predecessor estimate. Conservation bounds the defect of a possible extremizer; the equality case explains why the resulting law is close to the extremal Bernoulli distribution. After coarse localization, elementary inequalities place that law in one support region and count conditioning gives the strict comparison.

The small-noise estimate keeps its errors proportional to $\lambda$, which is essential near zero. The five-count estimate handles $1/100 \le \lambda \le 7/20$, where its uniform error is smaller than the crossing loss. At zero noise the law is Bernoulli and the strict finite-binomial theorem applies directly.

The argument excludes every attained nonnegative-excess pair at $n \ge 700$ when the sharp predecessor bound and the entry hypotheses hold. Consequently a proof of $C_j \le C_{\mathrm{E}}$ for all $1 \le j \le 699$ would complete the global sharp inequality. Those finite bounds are not supplied here. This work provides neither an independent four-summand proof nor a complete base through 41. The retained computational entry can be replayed separately from the analytical comparison and its short scalar checks; none proves the missing base.

**AI assistance.**

AI systems were used extensively for mathematical exploration, exact checking programs, source and bibliography comparisons, and drafting and revision. Separate AI review roles examined component arguments; their agreement is not a substitute for mathematical proof or external human review. Failed arguments remain diagnostics, not premises. The authors are responsible for the content and presentation.

**Sources.**

The immutable revision snapshot contains the revised proof, scalar checkers, and public execution records under `manuscript/compact-v16/publication/`. This snapshot precedes only the source-citation and release-metadata update and the corresponding rebuild; the mathematical arguments and checkers are unchanged. The retained entry certificate is also available at the original immutable baseline. Appendix E identifies the certificate, its exact arithmetic, and the commands for checking this revision. A document build checks presentation, not theorem validity.

# A A uniform Fourier enclosure with explicit input hypotheses

We prove the compensated Fourier formula and the complete low- and high-frequency estimates used in Section 5. Every assertion concerns an actual marginal with the displayed moment and event bounds. Attainment and acquisition of those bounds are not premises of the Fourier proof itself; their use is ordered in that section.

For each integer $n \geq N \geq 42$, let $X = X_n$ be a real random variable with $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, and $\mathbb{E}|X|^3 < \infty$. Neither the law nor the real raw threshold $H = H_n$ must be fixed as $n$ varies. Write

$$f(t) = \mathbb{E}e^{itX}, \quad \gamma = \mathbb{E}X^3, \quad e_n(t) = e^{-nt^2/2}(1 - in\gamma t^3/6), \quad V_n = f^n - e_n.$$

The two complete integrals are

$$I_0(H) = \int_0^{1/2} \frac{|\Im(e^{-itH}V_n(t))|}{t^3}\, dt, \qquad I_\infty = \int_{1/2}^{\infty} \frac{|V_n(t)|}{t^3}\, dt.$$

## A.1 The inversion edge, including atoms and moving thresholds

The sum law and its signed Gaussian approximation have the same mass and first moment. Subtracting these two terms before integration makes the Fourier expression absolutely integrable. The following elementary identity also explains why no continuity assumption at the threshold is needed.

**Lemma A.1.** *If a finite signed measure $\mu$ has mass and first moment zero and $\int x^2\, d|\mu|(x) < \infty$, then, for every $H \in \mathbb{R}$,*

$$\int (x - H)|x - H|\, d\mu(x) = -\frac{4}{\pi} \int_0^{\infty} \frac{\Im(e^{-itH}\widehat{\mu}(t))}{t^3}\, dt.$$

*The integral on the right is absolutely convergent.*

*Proof.* For real $y$, substitution and integration by parts give

$$\int_0^{\infty} \frac{ty - \sin(ty)}{t^3}\, dt = \frac{\pi}{4} y|y|.$$

The boundary terms vanish: near zero the numerator is $O(t^3)$, and at infinity it is $O(t)$. The integrand has the sign of $y$, since $v - \sin v \geq 0$ for $v \geq 0$. To calculate the constant, nonnegative Tonelli and elementary Laplace integration give

$$\int_0^{\infty} \frac{1 - \cos t}{t^2}\, dt = \int_0^{\infty} s\left(\frac{1}{s} - \frac{s}{s^2 + 1}\right) ds = \frac{\pi}{2}.$$

The absolute double integral of the compensated expression against $d|\mu|(x)$ equals

$$\frac{\pi}{4} \int (x - H)^2 d|\mu|(x) < \infty.$$

Absolute Fubini is therefore valid. The linear compensation disappears because $\int (x - H) d\mu = 0$, proving both assertions. No continuity of a CDF at $H$ is involved. □

Apply this to the sum law minus the scale-$\sqrt{n}$ pushforward of the signed density

$$\phi(x)\{1 + \gamma(x^3 - 3x)/(6\sqrt{n})\}.$$

This signed density has mass one, first moment zero and finite absolute second moment. Its transform after scaling is $e_n$. Gaussian integration by parts, using $q_z(x) = (x-z)|x-z|$ and $q_z''' = 4\delta_z$, yields

$$\varepsilon_n := \mathbb{E}q_{H/\sqrt{n}}(S_n/\sqrt{n}) - \mathbb{E}q_{H/\sqrt{n}}(Z) - \frac{2\gamma\phi(H/\sqrt{n})}{3\sqrt{n}} = -\frac{4}{\pi n}\int_0^\infty \frac{\Im(e^{-itH}V_n(t))}{t^3}\,dt.$$

Consequently the error $E_n = -n^{3/2}\varepsilon_n/2$ satisfies

$$|E_n| \le \frac{2\sqrt{n}}{\pi}\{I_0(H) + I_\infty\}. \tag{45}$$

This is an identity and bound for each $n, H$; it exchanges no limit in $n$ with an integral or a varying law.

### A.2 Low frequencies: Taylor cancellation and moment bounds

Near zero, the Gaussian approximation cancels the first terms of the characteristic function. The fourth- and fifth-order coefficients must still be bounded, and the sixth moment controls the remaining Taylor error. These higher moments are explicit hypotheses of the estimate; in its application they become available only after support has been bounded.

Assume common bounds

$$\mu_4 := \mathbb{E}X^4 \le M \le 111/10, \quad |\gamma| \le g \le 37/20, \quad \mu_6 \le M_6,$$

and put

$$a = (3-\mu_4)/24, \quad z = (10\gamma - \mathbb{E}X^5)/120, \quad |a| \le A, \quad |z| \le Z.$$

Here $A, Z, M_6$ are nonnegative common ceilings, not quantities inferred from bounded third moments. Define

$$E = M_6/720 + g^2/72 + (1/2 + g^2/144)^3/5, \qquad B = A + Z/2 + E/4,$$

$$a_0 = \frac{N-1}{2N}, \quad b_0 = \frac{45-M}{96}(1-2/N) > 0, \quad \lambda_N = \sqrt{\pi/2}\left(\frac{N}{N-1}\right)^{3/2}.$$

**Proposition A.2.** *For all the laws and thresholds above,*

$$I_0(H) \le \frac{L_0 + L_H|H|}{\sqrt{n}} + \frac{L_2}{n},$$

*where, for any $c_0 \ge \lambda_N$ and $c_1 \ge \lambda_N/2$,*

$$\begin{aligned} L_0 &= c_0 Z + c_1 A g + 251 g^3/17280, \\ L_H &= c_0 A + 251 g^2/4800, \\ L_2 &= E/(2a_0^2) + B^2/(2b_0^3). \end{aligned}$$

*Proof.* Let $h(t) = \exp(-t^2/2 - i\gamma t^3/6)$. Taylor's integral remainder for $e^{itX}$, followed by expectation, gives its expansion through degree five with error at most $\mu_6 t^6/720$. For $w = -t^2/2 - i\gamma t^3/6$ and $0 \le t \le 1/2$,

$$|w| \le t^2(1/2 + g^2/144) < 1/6, \quad |e^w - 1 - w - w^2/2| \le |w|^3 e^{|w|}/6 \le |w|^3/5.$$

The first bound follows from $\sqrt{1/4+y} \le 1/2 + y$; for the last use $e^{1/6} \le (1-1/6)^{-1} = 6/5$. Comparing the two polynomials gives

$$f - h = -at^4 - izt^5 + r_6(t), \qquad |r_6(t)| \le Et^6, \qquad |f-h| \le Bt^4. \tag{46}$$

The degree-six term in $w^2/2$ costs $g^2t^6/72$; it has not been discarded.

Four integrations of $1-\cos u \ge 0$ show $\cos u \le 1-u^2/2+u^4/24$ for every real $u$. Independence gives $\mathbb{E}(X-X')^4 = 2\mu_4+6$, and hence

$$|f(t)|^2 \le 1-t^2+(M+3)t^4/12 \le 1-(45-M)t^2/48.$$

Thus $|f|,|h| \le e^{-(45-M)t^2/96}$ throughout the low interval. Every point on their complex joining segment obeys the same bound. Taylor's integral formula for the polynomial $v \mapsto v^n$ proves

$$f^n-h^n = n(f-h)h^{n-1}+R_2, \quad |R_2| \le \tfrac{1}{2}n(n-1)B^2t^8e^{-(n-2)(45-M)t^2/96}.$$

The leading remainder in (46) and this power remainder cost at most $L_2/n$ after division by $t^3$ and integration: use $n-1 \ge 2a_0n$, $(n-2)(45-M)/96 \ge b_0n$, $n(n-1) \le n^2$, and $\int_0^\infty t^3e^{-At^2}dt = 1/(2A^2)$, $\int_0^\infty t^5e^{-At^2}dt = A^{-3}$.

For the remaining leading terms let $b=n-1$ and $v=b\gamma t^3/6$. Their imaginary part after rotation by $Ht$ is bounded by

$$ne^{-bt^2/2}\{(Z+A|H|)t^5+bAgt^7/6\}.$$

The Gaussian moments of orders two and four, after dividing by $t^3$, give $n\sqrt{\pi/2}b^{-3/2}(Z+A|H|+Ag/2)$. This is at most $\lambda_N(Z+A|H|+Ag/2)/\sqrt{n}$. Finally put $u=n\gamma t^3/6$. Direct sine and cosine remainders give

$$|\Im[e^{-itH}(h^n-e_n)]| \le e^{-nt^2/2}(|u|^3/6+|H|tu^2/2).$$

The Gaussian moments of orders six and four respectively give $\sqrt{\pi/2}g^3/(432\sqrt{n})$ times 5, and $\sqrt{\pi/2}g^2|H|/(24\sqrt{n})$. Using $\sqrt{\pi/2} < 251/200$ gives precisely the two displayed rational coefficients. Every extended integrand was nonnegative. The only index comparisons were the displayed rational inequalities; in particular the law and $H$ can vary arbitrarily with $n$. □

In the selected enclosure one has actual support $|X| \le R$ with $1 \le R \le 6$ and $\mathbb{E}|X|^5 \le E_5$. Then $M_6 = RE_5$ and $A = \max(2, M-3)/24$ are valid. Its sharper $Z$ is also licensed: for $Q = X^2-\gamma X-1$ direct multiplication gives

$$\mathbb{E}Q^2 = \mu_4-1-\gamma^2, \quad \mathbb{E}X^5 = \gamma^3+2\gamma+\mathbb{E}[Q^2(X+2\gamma)].$$

Reflect this auxiliary calculation if necessary so that $t=|\gamma| \ge 0$. For any $r \ge \max(R, 2g)$ the two signs of $\mathbb{E}X^5-10\gamma$ are at most

$$P_r(t) = r(M-1)+(10-2M)t-rt^2+t^3, \quad Q_r(t) = r(M-1)+(2M-10)t-rt^2-t^3.$$

The coefficients of $\mu_4$ before replacing it by $M$ are $r-2t, r+2t \ge 0$. If $M \le 5$, $P_r \ge Q_r$ and $P_r \le r(M-1)+(10-2M)t-(r-g)t^2$, whose maximum on $[0,g]$ is at the clipped vertex used in the code. If $M > 5$, $P_r \le r(M-1)$ and $Q_r \le r(M-1)+(2M-10)t-rt^2$, again with the stated clipped vertex. For $r=6$, $P_6'' = -12+6t < 0$ on $[0,g]$. If $M \le 5$ and $P_6'(g) \ge 0$, its maximum is exactly $P_6(g)$; otherwise the preceding quadratic gives $6(M-1)+(10-2M)^2/[4(6-g)]$. For $M > 5$ the unrestrained vertex bound is $6(M-1)+(2M-10)^2/24$. Taking the minimum of these proved ceilings and $E_5+10g$, then dividing by 120, gives a valid coefficient ceiling $Z$. The radius is an actual support input, not a free tuning parameter.

### A.3 High frequencies: six contributions and uniform index bounds

A nearly lattice law can retain characteristic-function peaks away from zero. The high-frequency argument therefore uses an event involving the actual difference $X - X'$ and partitions the oscillations around their zeros. The six contributions below cover intermediate frequencies, nonzero-cell cores, nonzero-cell wings, the small-angle part of the zero cell, the rest of the zero cell, and the signed Gaussian comparator. Each optional improvement has its own stated positivity and monotonicity conditions.

For independent copies of the same marginal suppose an actual event $\mathcal{E}$ satisfies

$$\mathbb{P}(\mathcal{E}) \geq \alpha > 0, \quad |X - X'| \geq d > 0 \text{ on } \mathcal{E}, \quad \mathbb{E}[(X - X')^2 \mid \mathcal{E}] \leq M_E.$$

Fix $b = 9/10$, $0 < r \leq 1/4$, and

$$h = 333/106 - (22/7)^3 r^2/6 > 0, \quad \kappa = \alpha(23/48)^2 d^2, \quad L(x) = \frac{x+2}{x^2+4x+2},$$

$$S_3 = \sum_{j=1}^{20} j^{-3} + 1/800, \quad S_5 = \sum_{j=1}^{20} (j-r)^{-5} + \frac{1}{4(20-r)^4}.$$

Let $c(u) = u/2 - (M+3)u^2/24$ and set

$$c_* = \min\{c(1/4), c(81/100)\}, \quad c_q = 1/8 - (M+3)/384, \quad s = [1 - (M+3)53/600]/2.$$

**Proposition A.3.** *Suppose $c_* > 0$. For every $n \geq N$, $\sqrt{n} I_\infty$ is at most the sum of the following six terms:*

$$\begin{aligned}
T_1(n) &= 2\sqrt{n} e^{-nc_*}, \\
T_2(n) &= \frac{M_E}{4\pi^{3/2} h \sqrt{\alpha}} \{S_3 + 3S_5/(n\alpha h^2)\}, \\
T_3(n) &= \frac{M_E \sqrt{n}}{2\pi^2} e^{-n\alpha h^2 r^2}, \\
T_4(n) &= \frac{e^{-n\kappa b^2}}{2b^4 \kappa \sqrt{n}}, \\
T_5(n) &= \frac{\sqrt{n}}{2b^2} e^{-n\alpha(23/48)^2}, \\
T_6(n) &= \sqrt{n} e^{-n/8} \{2L(n/8) + \sqrt{M-1}/3\}.
\end{aligned}$$

*If $s > 0$, $T_1$ may be replaced by its minimum with $2\sqrt{n} e^{-nc_q} L(ns/4)$. The fourth term may be replaced by its minimum with $\sqrt{n} e^{-n\kappa b^2} L(n\kappa b^2)/(2b^2)$. The common-floor sum at $N$ is valid if*

$$2Nc' > 1 \quad \text{for } c' \in \{c_*, \alpha h^2 r^2, \alpha(23/48)^2, 1/8\}.$$

*For the optional first term also require $2Nc_q > 1$; for the optional fourth term require $N\kappa b^2 \geq 1/4$. If an optional gate fails, the simpler term remains available.*

*Proof.* The triangle inequality separates $|f^n|$ and the entire comparator. On $[1/2, b]$, symmetrization gives $|f|^n \leq e^{-nc(t^2)}$. The function $c$ is concave, so $c(t^2) \geq c_*$ and the integral of $t^{-3}$ is less than two. This gives $T_1$. Also

$$c(u) - c_q - s(u - 1/4) = (M+3)(u - 1/4)(81/100 - u)/24 \geq 0.$$

When $s > 0$ this chord may be integrated to infinity.

For completeness, the tail calculation used here is, for $x = nab^2 > 0$,

$$\sqrt{n}\int_b^\infty e^{-nat^2}t^{-3}dt \le \frac{\sqrt{n}e^{-x}}{2b^2}L(x). \tag{47}$$

To see where the rational function $L$ comes from, let $U$ have density $e^{-u}$ on $u \ge 0$. Substitution followed by positive Fubini gives

$$\sqrt{n}\int_b^\infty e^{-nat^2}t^{-3}dt = \frac{\sqrt{n}e^{-x}}{2b^2}\mathbb{E}\frac{U}{x+U}.$$

For example, after putting $v = na(t^2 - b^2)$ the remaining factor is $\int_0^\infty xe^{-v}(x+v)^{-2}dv$; integration by parts makes it $\mathbb{E}[U/(x+U)]$. For $x > 0$ put

$$J_x = \mathbb{E}\frac{1}{x+U}, \qquad v_0 = \begin{pmatrix}1\\1\end{pmatrix}, \qquad G = \mathbb{E}\left[(x+U)\begin{pmatrix}1 & U\\ U & U^2\end{pmatrix}\right] = \begin{pmatrix}x+1 & x+2\\ x+2 & 2x+6\end{pmatrix}.$$

The entries use $\mathbb{E}U = 1$, $\mathbb{E}U^2 = 2$, and $\mathbb{E}U^3 = 6$. For $q(U) = c_1 + c_2U$, weighted Cauchy–Schwarz in its squared-residual form gives

$$0 \le \mathbb{E}\frac{[1-(x+U)q(U)]^2}{x+U} = J_x - 2v_0^Tc + c^TGc.$$

The matrix $G$ is positive definite: its leading entry is positive and $\det G = x^2 + 4x + 2 > 0$. Choosing $c = G^{-1}v_0$ yields

$$J_x \ge v_0^TG^{-1}v_0 = \frac{x+3}{x^2+4x+2}, \qquad \mathbb{E}\frac{U}{x+U} = 1 - xJ_x \le \frac{x+2}{x^2+4x+2} = L(x).$$

This proves (47) for every $a, b, n > 0$ with the same function $L$. Applying it at endpoint $1/2$ and rate $s$ proves the optional first term.

For $t \ge b$, symmetrization, the actual-event probability bound, and convex Jensen give

$$|f(t)|^n \le e^{-n\mathbb{E}\sin^2(t(X-X')/2)} \le \mathbb{E}[e^{-n\alpha\sin^2(tD/2)} \mid \mathcal{E}], \quad D = |X - X'|.$$

Positive Tonelli allows a separate partition for every $D > 0$. With $\ell = 2\pi/D$, use the zero cell $[0, \ell/2]$ and nonzero cells $\ell[j - 1/2, j + 1/2]$, $j \ge 1$. In a nonzero cell write $t = \ell(j + v)$. On its core $|v| \le r$, $|\sin \pi v| \ge h|v|$; on its wings $r \le |v| \le 1/2$ it is at least $hr$. These follow from $\sin x \ge x - x^3/6$ and monotonicity of sin on $[0, \pi/2]$, with directed bounds on $\pi$. For $|v| \le r$ Taylor's integral remainder gives

$$\{(j-v)^{-3} + (j+v)^{-3}\}/2 \le j^{-3} + 6(j-r)^{-5}v^2.$$

Extend the nonnegative Gaussian majorant over the whole real line. Its zeroth and second moments, the Jacobian $t^{-3}dt = D^2(4\pi^2)^{-1}(j+v)^{-3}dv$, and $\mathbb{E}[D^2 \mid \mathcal{E}] \le M_E$ give $T_2$. The two series tails are bounded by their decreasing integrals after 20. The total wing weight is at most $\int_{\ell/2}^\infty t^{-3}dt = D^2/(2\pi^2)$, giving $T_3$. Extending these sets below $b$ only increases their nonnegative bounds.

In the zero cell put $u = tD/2 \le \pi/2$. Where $u \le 1/2$, $\sin u \ge 23u/24$, and $D \ge d$ yields rate $\kappa$. Extending the set to $[b, \infty)$ gives $T_4$ from $t^{-3} \le t/b^4$; (47) gives its optional improvement. Where $u \ge 1/2$, monotonicity gives $\sin u \ge 23/48$ and the total weight is at most $1/(2b^2)$, proving $T_5$. Endpoints have zero Lebesgue measure even if the conditional distance distribution has atoms.

Finally $|e_n(t)| \le e^{-nt^2/2}(1 + n|\gamma|t^3/6)$. The first part on the entire $[1/2, \infty)$ uses (47); the second uses $\int_{1/2}^\infty e^{-nt^2/2}dt \le 2e^{-n/8}/n$. Pearson's identity $\mu_4 - 1 - \gamma^2 = \mathbb{E}(X^2 - \gamma X - 1)^2 \ge 0$ gives $T_6$. No comparator tail is dropped.

For floor evaluation, $\sqrt{n}e^{-nc'}$ decreases for $2nc' > 1$, whereas $n^{-1}$ and $n^{-1/2}e^{-nc'}$ decrease for every $c' > 0$. Also $L'(x) = -(x^2 + 4x + 6)/(x^2 + 4x + 2)^2 < 0$. Thus the optional first term decreases under its own gate, and the comparator decreases for $n \geq 5$. The sign of the derivative of $\sqrt{x}e^{-x}L(x)$ is the sign of

$$4 - 10x - 22x^2 - 13x^3 - 2x^4,$$

which is negative at $1/4$ and strictly decreasing on $(0, \infty)$. This proves the optional zero-cell gate. Every comparison is of a common scalar majorant, not of characteristic functions for different laws. □

### A.4 The precise closed Fourier conclusion

Under the preceding moment, actual-event and scalar gates, let $K_H$ be any complete HIGH sum evaluated at $N$, including its permitted optional improvements. Then

$$|E_n| \leq \frac{2}{\pi}\{L_0 + K_H + L_2/\sqrt{N} + L_H|H|\} \leq \frac{16}{25}\{L_0 + K_H + L_2/\sqrt{N} + L_H|H|\}. \tag{48}$$

The second inequality uses $\pi > 25/8$ and nonnegative coefficients. This is the constant Fourier budget and raw-threshold slope used in Section 5. Complete HIGH bounds for several valid windows may be minimized. The LOW coefficients must remain one complete valid triple.

For rational acceptance, positive denominator roots are rounded downward, numerator roots upward, and $\pi > 333/106$, $\sqrt{\pi} > 4431/2500$, $\pi < 22/7$ are directed correctly. These constants require no numerical transcendental oracle: polynomial division gives

$$22/7 - \pi = \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx > 0,$$

and the tangent addition formula with its first-quadrant branch gives

$$\pi = 16\arctan(1/5) - 4\arctan(1/239).$$

Four alternating terms for the first arctangent and the upper bound $\arctan(1/239) < 1/239$ give $\pi > 333/106 > 25/8$ by rational comparison. Squaring then verifies the two Gaussian constant enclosures used above. Its exponential helper replaces $x \geq 0$ by a downward rational $y \leq \min(x, 64)$ and returns an upward rounding of $1/\sum_{j=0}^{64} y^j/j!$. This is at least $e^{-x}$. The floor argument applies to the underlying analytic bounds, not to a claim that this rounded helper itself preserves strict monotonicity.

The estimates are uniform over all laws and thresholds satisfying the stated inequalities at the actual index. They require no monotonicity of extremal Berry–Esseen constants, no fixed-law asymptotic limit, and no unproved disappearance of a lattice tail.

## B Elementary Gaussian constants

**Lemma B.1** (Elementary Gaussian constants)**.** *For the standard normal density,*

$$\|\phi''\|_\infty = \phi(0) < 2/5, \qquad \|\phi'''\|_\infty \leq 2/\pi, \qquad 1/(4\pi) < 2/25.$$

*Moreover* $819/2000 < C_{\mathrm{E}} < 20487/50000 < 41/100$ *and* $\phi(0) > 0.39886 > 0.3988 = 997/2500$.

We give the exact comparisons used in Lemma B.1.

*Proof.* Since $\phi''(x) = (x^2-1)\phi(x)$, the norm on $x^2 \le 1$ is at most $\phi(0)$. On $x^2 \ge 1$, the maximum of $(x^2-1)e^{-x^2/2}$ is $2e^{-3/2} < 1$, attained when $x^2 = 3$. Thus the norm is exactly $\phi(0)$.

For completeness, the Gaussian transform can be obtained without an inversion theorem for the distribution being studied. Differentiating $J(t) = \int_{\mathbb{R}} e^{-x^2/2}e^{itx}\,dx$ under the integral and integrating by parts give $J'(t) = -tJ(t)$. The Gaussian integral, evaluated by squaring it and using polar coordinates, gives $J(0) = \sqrt{2\pi}$. Hence $J(t) = \sqrt{2\pi}e^{-t^2/2}$; applying the same formula with the variables interchanged gives

$$\phi(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-t^2/2}e^{-itx}\,dt.$$

Three differentiations are justified by the integrability of $|t|^3e^{-t^2/2}$. Consequently

$$\|\phi'''\|_\infty \le \frac{1}{2\pi}\int_{\mathbb{R}} |t|^3 e^{-t^2/2}\,dt = \frac{2}{\pi}.$$

The last integral equals four after the substitution $w = t^2/2$ on the positive half-line.

Here are elementary rational comparisons for all numerical constants used in this section. Put $a = \arctan(1/5)$ and $b = \arctan(1/239)$. Tangent addition gives $\tan(2a) = 5/12$, $\tan(4a) = 120/119$ and $\tan(4a-b) = 1$. Also

$$0 < 4(1/5 - 1/(3\cdot 5^3)) - 1/239 < 4a - b < 4/5 < 1 < \pi/2.$$

The arctangent inequalities follow by integrating $1 - t^2 < 1/(1+t^2) < 1$, while $\pi/2 = 2\arctan(1) > 1$ follows by integrating $1/(1+t^2) > 1/2$ on $[0,1)$. It follows that $\pi = 16a - 4b$ and

$$\pi > 16(1/5 - 1/(3\cdot 5^3)) - 4/239 = 281476/89625 > 25/8.$$

This proves $\phi(0) < 2/5$ and $1/(4\pi) < 2/25$.

To bound $C_{\mathrm{E}}$ more closely, define

$$s_k(x) = \sum_{j=0}^{k-1} \frac{(-1)^j x^{2j+1}}{2j+1}, \quad \pi_- = 16s_6(1/5) - 4s_1(1/239), \quad \pi_+ = 16s_5(1/5) - 4s_2(1/239).$$

The remainder in the finite geometric expansion of $1/(1+t^2)$ has its stated alternating sign, so integration gives $\pi_- < \pi < \pi_+$. Let $r_- = 3162277660/10^9$ and $r_+ = 3162277661/10^9$. Squaring shows $r_-^2 < 10 < r_+^2$. The following comparisons are exact rational ones:

$$\left(\frac{819}{2000}\right)^2 < \frac{19+6r_-}{72\pi_+} < C_{\mathrm{E}}^2 = \frac{19+6\sqrt{10}}{72\pi} < \frac{19+6r_+}{72\pi_-} < \left(\frac{20487}{50000}\right)^2.$$

For example, each outer comparison follows by multiplying by its positive rational denominator; the displayed finite sums specify those denominators completely. All quantities are positive, so taking square roots proves the claimed bounds. In particular, $0.4095 < C_{\mathrm{E}} < 0.40974 < 0.41$. The same upper endpoint for $\pi$ is smaller than $22/7$. Since $(0.39886)^2 < 7/44$, we also obtain $\phi(0) > 0.39886 > 0.3988$. □

**Lemma B.2** (The narrow entry enclosure)**.** *The target constant satisfies*

$$c_- := \frac{4097321837023}{10^{13}} < C_{\mathrm{E}} < \frac{4097321837025}{10^{13}} =: c_+.$$

*Proof.* Use the same finite sums $s_k$ and Machin identity proved above, now with

$$\pi_L = 16s_{10}(1/5) - 4s_5(1/239), \qquad \pi_U = 16s_9(1/5) - 4s_4(1/239).$$

An even number of terms bounds each arctangent from below and an odd number bounds it from above. The negative coefficient of the second arctangent reverses its choice of endpoint, so $0 < \pi_L < \pi < \pi_U$. Put

$$r_L = \frac{3162277660168379}{10^{15}}, \qquad r_U = \frac{3162277660168380}{10^{15}}.$$

Integer squaring gives $r_L^2 < 10 < r_U^2$. Direct rational comparisons give

$$c_-^2 < \frac{19 + 6r_L}{72\pi_U} < C_{\mathrm{E}}^2 < \frac{19 + 6r_U}{72\pi_L} < c_+^2.$$

The finite sums specify every rational in this display; each outer inequality can be checked by cross-multiplying positive integers. All terms are positive, so taking square roots gives the assertion. This directed enclosure, rather than the coarser bounds in the preceding lemma, supplies the precision used by the entry calculation. □

# C The identity without contact

This extension applies to every compactly supported standardized law, not just to an extremizer. Retain the definitions of $a, Q, h, \Delta, J, J_1, J_2, L$, and $\mathcal{R}_c$ from the main text, for arbitrary $H \in \mathbb{R}$ and $c > 0$. Introduce the contact residuals

$$\xi = \mathbb{E}[X\Delta(X)], \qquad R_{\mathrm{supp}} = \int_K \tau(t)\Delta(t)\,dt + \sum_{(u,v)} \tau_{u,v}(v-u)\Delta(v). \tag{49}$$

The first integral is with respect to Lebesgue measure on the closed support $K$, not to $P$. Since $\Delta$ is bounded on the convex hull and $\sum \tau_{u,v}(v-u) \le 1$, the series is absolutely convergent.

**Theorem C.1** (Conservation with contact residuals)**.** *For every such law, threshold, and positive coefficient,*

$$(n + H^2)a + nc\rho - \frac{n\varphi}{6}(\gamma + 3J) + n^{3/2}L = \mathcal{R}_c + n^{3/2}(R_{\mathrm{supp}} - H\xi). \tag{50}$$

*At a fixed-target maximizing pair, both residuals vanish.*

*Proof.* The gapwise identity in the proof of Theorem 4.1 holds without contact. Keeping its endpoint terms and the integral over $K$ adds exactly $R_{\mathrm{supp}}$ to the right side of (19). The zero-bias evaluation (20) is unchanged. In the first-moment calculation, $\mathbb{E}[Xh(X)] = \mathbb{E}[XQ(X)] - \xi$, so (21) becomes

$$M + \varphi = -\frac{\gamma H\varphi}{2n} + \frac{cK_1}{n} - \sqrt{n}\xi.$$

The same pointwise identity for $q_z$ therefore gives

$$\sqrt{n}\mathcal{A} = c\rho + a - \gamma\varphi/6 + P_0/n + \sqrt{n}\varepsilon/2 + H\sqrt{n}\xi,$$

with exactly the $P_0$ defined in the main proof. Equating the two expressions and multiplying by $n$ gives (50). At contact, $\Delta = 0$ on all of $K$, so every term in both residuals vanishes, including the Lebesgue integral. □

**Why the support integral is needed.**

Let $X$ be uniform on $[-\sqrt{3}, \sqrt{3}]$ and take $H = 0$. There are no support gaps, $F = T = 1/2$, $\beta = \gamma = 0$, and $K_q = 3\sqrt{3}/40$. Thus

$$R_{\text{supp}} = \int_K \tau \Delta = cK_q/n^{3/2} > 0.$$

Dropping the support integral would give a false identity even though all gap sums are empty. This example tests the extension without contact; it does not contradict the contact theorem. The leakage integral retains $[u, v)$ throughout: a right-endpoint atom belongs to $h(v)$, and a left-endpoint atom has weight zero.

# D Details of the coarse localization

This appendix proves the analytical estimates and specifies the finite conditions used by Theorem 5.1. The calculation has two coarse quadratic passes followed by four signed-moment passes. Each improvement is used only after its hypotheses have been established. Let $P$ be a centered, unit-variance law, $S_j$ a sum of $j$ independent copies, and set

$$\rho = \mathbb{E}|X|^3, \quad \beta = \mathbb{E}[X|X|], \quad \gamma = \mathbb{E}X^3, \quad c = C_{\text{E}} = \frac{3 + \sqrt{10}}{6\sqrt{2\pi}}.$$

The threshold $H$ always has the units of the raw sum. A fixed-target maximizer maximizes

$$a(P, H) = \sqrt{n}\{\mathbb{P}(S_n \le H) - \Phi(H/\sqrt{n})\} - c\rho.$$

We use Proposition 2.1 and its closed-support contact formula (5). In particular its closed-support contact equality, not ratio optimality, is the variational premise here. We consider a maximizer with $a \ge 0$. The actual normalized score is $c + a/\rho$; no upper bound on it is assumed.

## D.1 The first support bound

**Theorem D.1** (Initial moment and support entry)**.** *Assume $n \ge 42$, $C_{n-1} \le c$, and the universal structural inequality*

$$\sqrt{n}\,\Delta_n(P) \le \frac{3031}{10000}\left(\rho + \frac{323}{500}\right) \tag{51}$$

*for standardized laws with finite third moment. Then a fixed-target maximizer with $a \ge 0$ satisfies*

$$1 \le \rho < \frac{7361}{4000} < \frac{1841}{1000}, \qquad |\beta| < \frac{321}{500}, \qquad \operatorname{supp} P \subset (-96/25, 96/25). \tag{52}$$

*The theorem is uniform in all real $H$ and in both signs of a support point. It requires neither $C_{n-2}$ nor an excess bound of order $1/n$.*

The structural inequality (51) is the external iid theorem of Shevtsova [16]. Only this affine bound is needed for the displayed moment radius; it is not a consequence of the finite arithmetic in this section.

*Proof.* Since $c\rho \le \sqrt{n}\Delta_n(P)$ and $c > 819/2000$, elementary division gives

$$\rho < \frac{(3031/10000)(323/500)}{819/2000 - 3031/10000} = \frac{7361}{4000}.$$

The other affine bound $0.3322(\rho + 0.429)$ alone would give the larger number $712569/386500$; it must not be used to claim $7361/4000$. Lemma 2.4 gives

$$\rho \ge \frac{1+\beta^2}{\sqrt{1-\beta^2}}.$$

This function increases with $|\beta|$. For $r = 1841/1000$, $b = 321/500$,

$$(1+b^2)^2 - r^2(1-b^2) = \frac{93288849}{50000000000} > 0,$$

which proves the second assertion without assuming bounded support.

Write $v = n-1$, $s = \sqrt{n/v}$, $F_v(x) = \Phi(x/\sqrt{v})$ and $e_n = n(s-1) - 1/2$. Closed-support contact, the upper predecessor bound $F_{n-1,P}(H-t) \le F_v(H-t) + c\rho/\sqrt{v}$, and $\sqrt{n}(F_{n,P}(H) - \Phi(H/\sqrt{n})) = c\rho + a$ give

$$0 \le H_n(H,t) + c\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho e_n\} - na \quad (t \in \operatorname{supp} P), \tag{53}$$

where exactly

$$\frac{H_n(H,t)}{n^{3/2}} = F_v(H-t) - F_{v+1}(H) + tF'_{v+1}(H) - \frac{t^2-1}{2}F''_{v+1}(H).$$

The extra $-na$ can be discarded in this upper bound. This is the precise reason the predecessor assumption is $C_{n-1} \le c$, not a comparison with the actual score. No continuity assumption on the distribution function at $H-t$ is inserted.

Rationalization gives

$$ne_n = \frac{s^2(2s+1)}{2(s+1)^2}, \qquad s < 81/80 =: S, \qquad ne_n < \frac{793881}{2073680} < \frac{48}{125}.$$

The first expression increases for $s \ge 1$, as its logarithmic derivative is $2/s + 2/(2s+1) - 2/(s+1) > 0$. Thus $e_n < 8/875$.

For completeness we rederive the Gaussian estimate used at the radius. The heat identity $\partial_v F_v = F''_v/2$ permits spatial expansion of $F_v(H-t)$ to order four, variance expansion of $F_{v+1}$ to order two, and variance expansion of $F'_{v+1}, F''_{v+1}$ to order one. The cubic coefficient is $(-t^3+3t)/6$, and the quartic coefficient is $(t^4-6t^2+3)/24$. Therefore, for $t = \pm u$, $z_v = H/\sqrt{v}$,

$$H_n(H, \pm u)/s^3 = \pm A(1-z_v^2)\phi(z_v) + B\phi'''(z_v) + \mathcal{E}, \tag{54}$$

where $A = (u^3-3u)/6$, $H_4 = u^4 - 6u^2 + 3$ and $B = H_4/(24\sqrt{v})$. The four error bounds after division by $s^3$ are respectively

$$\frac{u^5\|\phi^{(4)}\|_\infty}{120v}, \quad \frac{\|\phi^{(5)}\|_\infty}{48v^{3/2}}, \quad \frac{u\|\phi^{(4)}\|_\infty}{8v}, \quad \frac{|u^2-1|\|\phi^{(5)}\|_\infty}{16v^{3/2}}.$$

Gaussian Fourier inversion gives norms at most $3\phi(0) < 6/5$ and $8/\pi < 51/20$. Since $v \ge 41$ and $\sqrt{v} > 6$, for $u \ge 1$ this gives

$$|\mathcal{E}| < E(u) := u^5/4100 + 3u/820 + 17(3u^2-2)/78720.$$

These errors are uniform in $H$; they use no higher moment of $P$.

If $A, B > 0$ and $B \le A/4$, the leading part of (54) is at most $\phi(0)(A + 9B^2/(4A))$ for both signs. Indeed, for the positive sign and $|z_v| \le 1$, it is at most $\phi(0)(A - Az_v^2 + 3B|z_v|)$, and complete the square. Outside that interval the first term is nonpositive. For the negative sign the first term is at most $2e^{-3/2}A\phi(0) < A\phi(0)/2$. In both cases use $\|\phi'''\|_\infty < 2\phi(0)$: on $[0, \sqrt{3}]$ the polynomial $3x - x^3$ is at most two, and beyond $\sqrt{3}$ use $x^3 e^{-x^2/2} \le 3\sqrt{3}e^{-3/2} < 2$.

For frozen $r, b$ above define

$$Q(u) = -u^3 + \tfrac{3}{2}ru^2 + 3bu + (8/875)r,$$
$$P(u) = \tfrac{2}{5}S^3\left(A^2 + \frac{H_4^2}{256 \cdot 41}\right) + S^3E(u)A + \frac{819}{2000}Q(u)A.$$

At $R = 96/25 < 6$ we have $A > 0$, $0 < H_4 < 36A$, hence $B < A/4$. The sign $Q(R) < 0$ is checked before replacing $c$ by its lower endpoint. The upper bound in (53), multiplied by $A$, is then strictly below $P(R)$, and exact arithmetic gives

$$-P(R) = \frac{2214258283536204102723}{10250000000000000000000} > \frac{1}{5}.$$

To cover each entire exterior ray, differentiation of the exact Gaussian expression gives

$$|\partial_t H_n(H, t)| < \tfrac{2}{5}(t^2/2 + 1/2 + e_n).$$

For verification put $y = (H - t)/\sqrt{n}$, $z = H/\sqrt{n}$; the derivative is $n\phi(z) + Ht\phi(z) - ns\phi(sy)$. The second-order Taylor error between $\phi(y)$ and $\phi(z)$ costs $\phi(0)t^2/2$, while $\partial_s[s\phi(sy)] = -\phi''(sy)$ costs $\phi(0)n(s-1)$. Define $L(u) = \frac{2}{5}(u^2/2 + 1/2 + 8/875) + (819/2000)Q'(u)$. At $R$, the six numbers $Q, Q', Q'', L, L', L''$ are strictly negative. Since $Q''' = -6$ and $L'' = 2/5 - 6(819/2000) < 0$, all their required signs persist for $u \ge R$. Thus both radial upper bounds decrease from a strictly negative boundary value. The contact inequality excludes every support point with $|t| \ge R$. □

## D.2 Higher moments and an actual Fourier event

Higher moments are used only after Theorem D.1. For $x = |X|$ the nonnegative measure $d\mu_0 = (x-1)^2(x + 1/2)\, dP$ has mass $\rho - 1$. Direct expansion gives

$$M_4 - 1 = \int \frac{(x+1)^2}{x + 1/2}\, d\mu_0,$$
$$M_5 - 1 = \int \frac{x^3 + 2x^2 + 3x + 3/2}{x + 1/2}\, d\mu_0,$$

where $M_j = \mathbb{E}|X|^j$. Both ratios increase for $x \ge 0$: they equal $x + 3/2 + 1/[4(x + 1/2)]$ and $x^2 + 3x/2 + 9/4 + 3/[8(x + 1/2)]$, whose derivatives are nonnegative. Writing their values at $R = 96/25$ as $k_4(R), k_5(R)$, we obtain

$$M_4 \le \min(R\rho, 1 + k_4(R)(\rho - 1)), \quad M_5 \le \min(RM_4, 1 + k_5(R)(\rho - 1)), \quad M_6 \le RM_5. \tag{55}$$

Also $\gamma^2 \le M_4 - 1$, by expanding $\mathbb{E}(X^2 - 1 - \gamma X)^2 \ge 0$.

There is no need to assume a central support gap before the first Fourier bound. For independent copies let $E = \{(X - X')^2 \ge 9/25\}$ and let $M$ be any valid fourth-moment ceiling. Then

$$\mathbb{P}(E) \ge \alpha := \frac{(2 - 9/25)^2}{2M + 6} > 0, \quad |X - X'| \ge 3/5 \text{ on } E, \quad \mathbb{E}[(X - X')^2 \mid E] \le 2/\alpha. \tag{56}$$

Indeed $\mathbb{E}[(X-X')^2;E] \geq 2-9/25$ and $\mathbb{E}(X-X')^4 = 2M_4+6 \leq 2M+6$, so Cauchy proves the mass bound. The conditional second-moment bound follows from its unconditional value two. This is an event under one actual iid marginal, not a replacement characteristic function.

The same support proof will be used again on a smaller closed moment cell. Freeze upper bounds $r, b$ for $\rho, |\beta|$ in its polynomial $Q$, and test a proposed radius before changing any moment bound. For $R \in [5/2, 96/25]$, the gates $A > 0$, $0 < H_4 < 36A$, $Q(R) < 0$, $P(R) < 0$, and the six negative ray signs suffice. Their strictness excludes the boundary atoms as well as the open exterior rays. A radius that fails a gate is not used. The selected finite search is $R = 2.50, 2.51, \ldots, 3.84$; it only chooses among proved implications.

### D.3 The Fourier enclosure used by every pass

Let $f(t) = \mathbb{E}e^{itX}$ and $e_n(t) = e^{-nt^2/2}(1 - in\gamma t^3/6)$. For the remainder in (16), the compensated inversion proved in Appendix A gives

$$|E_n| \leq \frac{2\sqrt{n}}{\pi}\left\{\int_0^{1/2} \frac{|\Im[e^{-itH}(f(t)^n - e_n(t))]|}{t^3}dt + \int_{1/2}^{\infty} \frac{|f(t)^n - e_n(t)|}{t^3}dt\right\}.$$

The comparator is signed, has the correct mass and first moment, and its entire high-frequency tail is included. This formula applies at atoms and for arbitrary real $H$.

For clarity, the exact input and output of the selected Fourier step are as follows. Acquire $|X| \leq R$ first, and choose common ceilings $M \leq 11.1$, $g \leq 1.85$, $M_6$ for $M_4, |\gamma|, \mathbb{E}X^6$, together with

$$|(3 - M_4)/24| \leq A_4, \qquad |(10\gamma - \mathbb{E}X^5)/120| \leq Z_5.$$

Put

$$\begin{gathered} E_6 = M_6/720 + g^2/72 + (1/2 + g^2/144)^3/5, \\ B_6 = A_4 + Z_5/2 + E_6/4, \\ a_0 = (N-1)/(2N), \quad b_0 = (45 - M)(1 - 2/N)/96, \\ c_0 \geq \sqrt{\pi/2}(N/(N-1))^{3/2}, \qquad c_1 \geq c_0/2. \end{gathered}$$

A complete admissible low-frequency triple is

$$\begin{aligned} L_0 &= c_0 Z_5 + c_1 A_4 g + 251 g^3/17280, \\ L_H &= c_0 A_4 + 251 g^2/4800, \\ L_2 &= E_6/(2a_0^2) + B_6^2/(2b_0^3). \end{aligned}$$

One may take $A_4 = \max(2, M-3)/24$ and $M_6 = RM_5$. The sharper fifth-order coefficient used by the finite cover is proved in Appendix A; it follows from the exact identity for $X^2 - \gamma X - 1$, not from an assumed fifth moment.

The high-frequency input is an actual event for two iid copies:

$$\mathbb{P}(\mathcal{E}) \geq \alpha > 0, \qquad |X - X'| \geq d_E > 0 \text{ on } \mathcal{E}, \qquad \mathbb{E}[(X-X')^2 \mid \mathcal{E}] \leq M_E.$$

Proposition A.3 supplies a complete six-term bound $K_H$ at $N = 42$, subject to its explicit positive damping and all-index gates. The first event is (56); an opposite-sign event is used only after an actual central gap is acquired below. Different valid windows may be minimized only after all six terms, including the comparator, are present. No characteristic function of an artificial binary law is substituted for $f$.

The exact scalar output is

$$|E_n| \le K_F + s_F|H|, \qquad K_F = \frac{16}{25}(L_0 + K_H + L_2/\sqrt{N}), \quad s_F = \frac{16}{25}L_H. \tag{57}$$

All square roots and exponential bounds in the finite evaluation are directed outward. The proof in the appendix applies to every actual $n \ge N$ and allows the law and threshold to vary with $n$. Thus no fixed-law limit is being used as a uniform bound.

### D.4 Coarse coefficients before a central gap is known

Write the non-Fourier constant charge in (16) as

$$A_c^{\mathrm{law}} = \frac{3}{2}\beta(J+\gamma) + 3(\rho J_1 - J_2) + \frac{\rho}{2}M_4 - \frac{1}{4}M_5 - \frac{\rho}{2}.$$

Here $A_c^{\mathrm{law}}$ does not depend on $c$; the subscript records its role as the coefficient of $c$. We will bound it from above, retaining all signs in the later passes.

The following gap inequalities need no gap around zero:

$$-M_4/6 \le J_1 \le M_4/2, \qquad -3J_2 - M_5/4 \le 0, \qquad J \le \rho + 1. \tag{58}$$

For a gap $(u, v)$ of length $h$, the first pair follows from

$$\int_u^v (t-u)t\,dt + \frac{1}{2}\int_u^v t^2 dt = hv^2/2, \quad \frac{3}{2}\int_u^v t^2 dt - \int_u^v (t-u)t\,dt = h(2u+v)^2/6.$$

Sum with the nonnegative gap weights and use $\int t^2\tau = M_4/3$. On a negative gap, reflection and

$$\frac{1}{3}\int_u^v x^3 dx - \int_u^v (v-x)x^2 dx = u^3(v-u)/3 \ge 0 \quad (0 \le u \le v)$$

prove the second assertion; the positive portion only improves it. The last bound follows from $J \le \rho + 2m$ and $2m \le 1$.

For $1 \le r \le \rho$ and $R \le 6$, choose $\Gamma(R, r) \ge 0$ with

$$\Gamma(R,r) \ge \max_{r \le v \le R} \left\{ -\frac{v^2}{4} + \frac{(R+r)v}{3} - \frac{Rr}{2} + \frac{r^3(2R-r)}{12v^2} \right\}_+ . \tag{59}$$

Then

$$3(\rho J_1 - J_2) \le \frac{3}{2}\Gamma(R,r)(\rho - \gamma) + \frac{3}{8}\rho^2 J. \tag{60}$$

On the negative portion of a gap discard $|t| < \rho$, where its charge is nonpositive, and put $x = -t$. Since $1 \le r \le \rho$ and $|X| \le R$, the remaining disjoint contributions are bounded by

$$\begin{aligned} 3\int_r^R \tau(-x)(R-x)x(x-r)\,dx &= 3\mathbb{E}\left[X_-\mathbf{1}_{\{X_- > r\}}\int_r^{X_-}(R-x)x(x-r)\,dx\right] \\ &\le 3\Gamma(R,r)\mathbb{E}X_-^3. \end{aligned}$$

Nonnegativity permits the countable gap sum and these integrals to be interchanged by Tonelli. For $v \in [r, R]$, the inner integral divided by $v^2$ is exactly the expression in (59). Since $\mathbb{E}X_-^3 = (\rho - \gamma)/2$, this gives the first term of (60). On the positive portion, $t(\rho - t) \le \rho^2/4$ and $\sum \tau_{u,v}\int_u^v (t-u)dt = J/2$ give the other term. This argument includes gaps crossing zero and atoms at either endpoint.

The finite bound for $\Gamma$ partitions $[r, R]$ into 64 closed intervals. On each, maximize the concave quadratic at its clipped vertex $2(R+r)/3$ and its nonnegative reciprocal term at the left

endpoint. The alternative ceiling $5/2$ follows at $r=1, R=6$ from $3v^4-28v^3+66v^2-11>0$ on $[1,6]$: the endpoint values are $30, 205$ and the two interior critical values are $(85\mp 35\sqrt{5})/2>0$. The positive integral representation is increasing in $R$ and decreasing in $r$, so that endpoint choice encloses the present domain.

The elementary square

$$V=\mathbb{E}[|X|^3(|X|-\rho)^2]=M_5-2\rho M_4+\rho^3\geq 0$$

gives $\rho M_4/2-M_5/4\leq \rho^3/4$. Thus on $\rho\in[l,u]$, $|\beta|\leq b$, $M_4\leq M$, and with $0\leq \rho-\gamma\leq\Delta$, two complete constant bounds are

$$\begin{aligned} A_1(\Delta)&=\tfrac{3}{2}b(2u+1)+\tfrac{3}{2}\Gamma(R,l)\Delta+\tfrac{3}{8}u^2(u+1)+u^3/4-l/2,\\ A_2&=\tfrac{3}{2}b(2u+1)+2uM-l/2. \end{aligned}\tag{61}$$

Use $A=\min(A_1,A_2)$. The first uses (60); the second uses (58). They are alternative complete bounds, not partial costs to be mixed.

For the linear term $K_1$, the following three bounds are valid:

$$\begin{aligned} K_{\mathrm{rad}}&=3ub+\frac{R(R+2-3l/2)}{R+1/2}(u-1),\\ K_{\mathrm{univ}}&=6u-3u^2/2+3b,\\ K_R&=Ru/2+\max_{r\in\{l,u\}}|R/2-3r/2|g+3b. \end{aligned}$$

For the first use the positive measure $d\mu_0=(|X|-1)^2(|X|+1/2)dP$, of mass $\rho-1$, and the exact identity

$$K_1+3\rho\beta=\int \operatorname{sgn}(X)\frac{x(x+2-3\rho/2)}{x+1/2}\,d\mu_0,\qquad x=|X|.$$

The ratio is convex in $x\geq 0$ and decreases in $\rho$. Its negative part is at most $31/40$ on $\rho\leq 1.85$; require its displayed endpoint ceiling to exceed that number. The second bound uses $|\mathbb{E}[X|X|^3]-3\gamma|\leq 3\rho$ for $R\leq 6$, and $\rho\leq 2$. For the third use the signwise fourth-moment bounds $0\leq \mathbb{E}X_\pm^4\leq R(\rho\pm\gamma)/2$. Finally (58) gives

$$|\varphi\Omega|\leq v_H:=\frac{2}{5}\max\{1/2,2M/3-1/2\},\qquad \Omega=J_1+(M_4-3)/6.$$

**Lemma D.2** (A stronger signed fourth-moment charge)**.** *Let $|X|\leq R$, $\rho\in[l,u]$, $|\beta|\leq b<1$. Put $h_*=1+b$ and $a(r)=(3r-R)/2$. A valid bound for $|K_1|$ is the maximum on $[l,u]$ of*

$$B_R(r,b)=Rr-\tfrac{3}{2}r^2+3b+\tfrac{1}{2}(1+b)a(r)^2.$$

*If the three gates*

$$h_*\geq a(u),\qquad 3+2a(l)h_*-\tfrac{3}{2}h_*^2\geq 0,\qquad R-3u+\tfrac{3}{2}h_*^2\geq 0$$

*hold, it may be intersected with the smaller bound*

$$Ru-\tfrac{3}{2}u^2-3+3h_*+a(u)h_*^2-h_*^3/2.$$

*Proof.* For $\epsilon=\pm1$ set $h=1-\epsilon\beta>0$ and $\delta=\rho-\epsilon\gamma$. Signwise Cauchy and $m\leq 1/2$ give $\delta\geq h^2$. The fourth moments on the two signs give

$$\epsilon K_1\leq R\rho-\tfrac{3}{2}\rho^2-3+3h+a(\rho)\delta-\delta^2/(2h).$$

Completing the square in unrestricted $\delta$ and using $h \le h_*$ proves the first bound. With $\delta \ge h^2$, the last three terms maximize to

$$\psi_a(h) = \begin{cases} h(3 + a^2/2), & 0 < h \le a, \\ 3h + ah^2 - h^3/2, & h \ge a. \end{cases}$$

The first branch is increasing. The second has derivative $3 + 2ah - 3h^2/2$, a concave quadratic whose endpoint values on $[0, h_*]$ are nonnegative under the second gate. The branches agree at $h = a$; the first gate places $h_*$ in the second branch. The remaining derivative in $\rho$ is $R - 3\rho + 3h_*^2/2$, nonnegative by the third gate. This proves the claim, including all endpoints. The quadratic maximum in the first bound is evaluated at its endpoints and, when present, its concave vertex. □

Take the minimum of the applicable bounds for $|K_1|$ and call it K. No assertion about the signs of $\beta\gamma$ or $K_1\gamma$ is used.

### D.5 Two quadratic passes and the first actual gap

Use the 24 closed cells with endpoints

$$r_i = 1 + 3361i/96000, \quad i = 0, \dots, 24, \qquad N = 42,$$

and directed constants

$$c_- = 0.4097321837023 < C_{\mathrm{E}} < c_+ = 0.4097321837025,$$
$$\phi_l = 997/2500 < \phi(0), \quad \sigma_l = 3.162277660 < \sqrt{10}.$$

The Gaussian constants follow from the rational Machin estimates in Lemma B.1 and Appendix A; the narrow $c$ enclosure has the explicit directed recipe in Lemma B.2. The interval encloses the fixed coefficient c, not the actual normalized score.

On a cell $[l, u]$, form the moments and the universal actual event above, and compute the complete Fourier pair $K_F, s_F$. In pass 1 use $\Delta = 2u$, $A = \min(A_1(2u), A_2)$, and, for $t \in [c_-, c_+]$,

$$f(t) = tK + v_H + s_F, \qquad q(t) = tl - g/5.$$

Require $q > 0$. The bound $\mathcal{R}_c \le K_F + cA + f(c)|H| - q(c)H^2$ is affine in c at each fixed $|H|$, so its maximum is at an endpoint before completing squares. Let an overline denote upward rounding to $10^{-10}$. Compute

$$\begin{aligned} T_1 &= \overline{\max_{t=c_-,c_+} \{K_F + tA + f(t)^2/(4q(t))\}}, \\ \eta_1 &= (c_- l - T_1/N)/2, \\ U_1 &= \min\left\{T_1, \overline{\max_{t=c_-,c_+} \{K_F + tA + f(t)^2/[4(q(t) + \eta_1)]\}}\right\}, \\ d_1 &= \overline{6U_1/(N\phi_l)}. \end{aligned} \tag{62}$$

The strict gate $0 \le T_1 < Nc_- l$ is checked before $\eta_1$ or the resource bound is used. Corollary 4.2 then proves $\mathcal{R}_c \le T_1$, $\mathcal{U} \le U_1$, and $nD \le Nd_1$. The scalar bounds hold at every actual $n \ge N$.

For the second pass the circle in Lemma 3.1 gives

$$D \ge \sigma\rho - \gamma - 3\sqrt{\rho^2 - \gamma^2}, \qquad \sigma = \sqrt{10}.$$

Put $a_0 = \sigma_l - d_1/l$. A valid lower bound for $\gamma/\rho$ is

$$Y(a_0) = \begin{cases} -1, & a_0 \le -1, \\ (a_0 - 3\sqrt{10 - a_0^2})/10, & -1 < a_0 \le \sqrt{10}. \end{cases} \tag{63}$$

Indeed $y + 3\sqrt{1-y^2}$ increases from $-1$ to $\sqrt{10}$ on $[-1, 1/\sqrt{10}]$. Solving its level equation on that branch gives the smaller quadratic root in (63); it has $a_0 - y \ge 0$, so squaring introduced no false branch. Every y below it violates the required lower level. A downward rational bound $y_l$ is obtained by rounding the square root upward. Thus

$$g_l = \max\{-g, \sigma_l l - 3 - d_1, \min(l y_l, u y_l)\} \le \gamma,$$
$$\Delta = \min\{2u, (1-\sigma_l)l + 3 + d_1, u - g_l\}, \qquad \phi_- = \phi_l(1 - T_1/(Nc_- l)) > 0,$$
$$A = \min\{A_1(2u), A_1(\Delta), A_2\},$$
$$q(t) = \max\{tl - g/5, tl + \tfrac{1}{2}\min(\phi_- g_l, (2/5) g_l)\}.$$

The first two bounds also use $2m \le 1$. Each value of $q(t)$ is affine in t, since its two alternatives have the same slope l. Repeat the raw bound first, intersect it with $T_1$, then price phase using this new raw bound, intersect the resource with $U_1$, and finally retain $d_2 \le d_1$. A resource bound is never substituted for a raw bound in the density or phase license.

**Certificate condition D.3** (The two coarse passes)**.** The preceding coefficient domains, Fourier gates, endpoint enclosures and strict raw gates hold on every one of the 24 closed cells. Outward evaluation gives the following ceilings, and in every second-pass cell $d_2 < (\sigma_l - 1)l$:

| pass | $\max T$ | $\max U$ | $\max d$ | strict gap gates |
|---|---|---|---|---|
| 1 | 13.2818901598 | 10.9330610254 | 3.9164138937 | 23 |
| 2 | 11.5164967977 | 9.9790315907 | 3.5746638454 | 24. |

The minimum second-pass gap margin is at least $526024602543/1600000000000 > 0$.

The stated condition is a finite exact arithmetic obligation. The named two-pass proof object records and rechecks these inequalities; it is kept distinct from the analytic implications just proved. Although its implementation retains additional previously valid minima, the fresh quadratic ceilings attain all the displayed minima in this cover, so no earlier resource-refinement schedule is a mathematical premise here.

If zero belonged to the closed support, every gap would be on one side of zero and $J \le \rho$. Hence $D \ge (\sigma - 1)\rho$, contradicting the strict second-pass test. Centering and positive variance therefore give an actual central gap $(-A, B)$, with positive endpoints. This acquires an event, rather than supposing an artificial two-point kernel.

## D.6 Signed moments and physical gap costs

Write $\mu_4 = M_4$, $\eta_5 = M_5$, $\sigma = \sqrt{10}$. Put $a_\pm = (1 \pm \beta)/2$, $b_\pm = (\rho \pm \gamma)/2$. Both signs have positive probability and positive first moment, since a centered variance-one law cannot be supported on one half-line. Thus $|\beta| < 1$ and the signwise denominators below are positive whenever used; degenerate correction denominators are handled separately. Cauchy gives $m^2 \le a_+ a_-$ and $b_\pm m \ge a_\pm^2$. Consequently

$$\frac{(1+\beta)^{3/2}}{\sqrt{1-\beta}} - \rho \le \gamma \le \rho - \frac{(1-\beta)^{3/2}}{\sqrt{1+\beta}}, \qquad m \ge \max\left\{\frac{1+\beta^2}{2\rho}, \frac{a_+^2}{b_+}, \frac{a_-^2}{b_-}\right\}.$$

These statements also hold with a zero atom. Projection of $X|X|$ onto the span of $X$ and $|X|$ gives

$$\mu_4 \geq \rho^2 + \frac{(\gamma - \rho\beta)^2}{1 - \beta^2}.$$

For completeness, this follows by minimizing the nonnegative expectation $\mathbb{E}(X|X| - uX - v|X|)^2$ over $u, v$; its Gram matrix has diagonal entries one and off-diagonal entry $\beta$, and is positive definite.

Two support-localizing moment matrices give the additional upper bounds as follows. If $D_\pm = Ra_\pm - b_\pm > 0$, then

$$2m + \sum_{\pm} \frac{(Rm - a_\pm)^2}{D_\pm} \leq R, \qquad \mu_4 \leq R\rho - \sum_{\pm} \frac{D_\pm^2}{Rm - a_\pm}.$$

Indeed the first follows by summing the determinant inequalities for

$$\mathbb{E}[(R - |X|)(1, |X|)^{\mathsf{T}}(1, |X|); \pm X > 0].$$

The second follows from the corresponding matrices with an additional factor $|X|$. One more factor $|X|$ yields, sign by sign,

$$\mathbb{E}[X_\pm^5] \leq R\mathbb{E}[X_\pm^4] - \frac{(Rb_\pm - \mathbb{E}[X_\pm^4])^2}{D_\pm}.$$

Zero denominators are treated by dropping their nonnegative correction; they never imply exclusion without a separate contradictory numerator. The common $m$ is retained throughout.

A second useful upper bound for $m$ follows by integrating $(x - a)^2(R - x) \geq 0$ for $x = |X| \in [0, R]$. For every fixed $a > 0$,

$$2m \leq \frac{R - \rho + 2a + Ra^2}{a(2R + a)}.$$

The cover uses only the eleven rational choices $a = 2/10, \ldots, 12/10$.

Write $\delta = \gamma - \rho\beta$. Cauchy and a square completion on each sign give the signed constant bound

$$\frac{\rho}{2}\mu_4 - \frac{1}{4}\eta_5 \leq \frac{\rho^3}{4} - \frac{(\rho + \gamma)(\delta_+)^2}{8(1 + \beta)^2} - \frac{(\rho - \gamma)((-\delta)_+)^2}{8(1 - \beta)^2}.$$

Indeed on either sign, write $a = \mathbb{E}X_s^2$, $b = \mathbb{E}X_s^3$, $c = \mathbb{E}X_s^4$, $f = \mathbb{E}X_s^5$. Cauchy gives $c \geq b^2/a$, $f \geq c^2/b$, and $\rho c/2 - f/4 \leq \rho^2 b/4 - (c - \rho b)^2/(4b)$. Distance to the half-line $c \geq b^2/a$ proves the stated penalty on summing both signs. Independently, $\mu_4^2 \leq \rho\eta_5$ and the projection bound above give

$$\frac{\rho}{2}\mu_4 - \frac{1}{4}\eta_5 \leq \frac{\rho^3}{4} - \frac{(\gamma - \rho\beta)^4}{4\rho(1 - \beta^2)^2}.$$

Each is a complete deduction from the same square. There is a third independent deduction which is often stronger. For $Y = |X|$ and $\delta = \gamma - \rho\beta$, put

$$V = \mathbb{E}[Y(Y^2 - \rho Y)^2] = \eta_5 - 2\rho\mu_4 + \rho^3.$$

Since $\delta = \mathbb{E}[Y \operatorname{sgn}(X)(Y^2 - \rho Y)]$, weighted Cauchy gives $\delta^2 \leq 2mV$. Hence

$$\frac{\rho}{2}\mu_4 - \frac{1}{4}\eta_5 \leq \frac{\rho^3}{4} - \frac{\delta^2}{8m}. \tag{64}$$

The maximum of these three separately valid losses can be subtracted. Their sum cannot. A zero atom contributes zero to the weighted measure.

**An actual central gap without assuming $D < 2$**

The physical decomposition (13) shows that $D < (\sigma-1)\rho$ implies an actual central gap. Use its same nonnegative terms, writing $T = T_+ + T_-$ and $Q = (\sqrt{\sigma+1}A - \sqrt{\sigma-1}B)^2$. Hence $T \le D/(\sigma-1)$. The additional inequality $J \le \rho + 2mAB$ gives $6mAB \ge (\sigma-1)\rho - D$. Consequently, if $\rho \ge l$, $D \le d < (\sigma_l - 1)l$ and $m \le m_u$,

$$(A+B)^2 \ge \frac{l - d/(\sigma_l - 1) + \{(\sigma_l-1)l - d\}/3}{m_u} > 0. \tag{65}$$

This is the distance lower bound used in the complete high-frequency event. It uses the actual moment m and the actual support endpoints.

Moreover $A \le a_-/m$, $B \le a_+/m$ and $J \le \rho + 2m$. The last bound gives the compulsory defect

$$D \ge \sigma\rho - \gamma - 6m.$$

The inequalities $T - J_o \ge 2BR_+ + 2AR_-$ give the useful additional endpoint constraint

$$D \ge (\sigma-1)b_+ + 6a_+B - (\sigma+5)mB^2.$$

If $D < (\sigma-1)b_+$, this downward quadratic forces $B$ to be at least its positive root. The reflected constraint has $a_-, b_-, A$ and the coefficients $\sigma+1, \sigma+7$. In a closed moment box, positive coefficients are rounded down, negative coefficients up, and the lower square-root endpoint is used in the resulting root bound.

The actual opposite-sign event has probability $2pq$. There is no atom at zero. Cauchy gives the necessary probability interval $2m_-^2/(1+\beta_+) \le p \le 1 - 2m_-^2/(1-\beta_-)$ on a moment cell. In particular, $2pq \ge (1+\beta^2)^2/(2\rho^2)$, using $m \ge (1+\beta^2)/(2\rho)$ and $pq \ge m^2$. The minimum of $p(1-p)$ on a closed interval is attained at an endpoint.

For independent copies one has exactly

$$\mathbb{E}[(X-X')^2 \mid XX' < 0] = \frac{1+\beta(1-2p)+4m^2}{2pq} = \frac{1+4m^2+\beta}{2p} + \frac{1+4m^2-\beta}{2q}.$$

For an upper bound, use the upper $m$ endpoint and both endpoints of $p$ and $\beta$. The expression is convex in $p$ and affine in $\beta$. This improves the former $2/\mathbb{P}(XX' < 0)$ bound inside the complete high-frequency estimate. Both copies still have the same actual marginal.

**Preserving central and outer integral costs**

Let

$$P_1(A,B) = B^3/3 + AB^2/2 - A^3/6, \quad P_2(A,B) = B^4/4 + AB^3/3 - A^4/12, \quad G = \rho P_1 - P_2.$$

Then $J_{1,c} = mP_1$, $J_{2,c} = mP_2$. Since $\partial_B G = B(A+B)(\rho - B)$,

$$G(A,B;\rho) \le G(A,\rho;\rho) = \rho^4/12 + \rho^3 A/6 - \rho A^3/6 + A^4/12.$$

The last expression increases with $A$ because its derivative is $(\rho - A)^2(\rho + 2A)/6$, and is convex in $\rho$ because its second $\rho$ derivative is $\rho^2 + \rho A$. Its closed-rectangle maximum is therefore bounded using the upper $A$ endpoint and both $\rho$ endpoints.

Positive outer gaps satisfy $(t-a)(\rho - t) \le (\rho - B)_+^2/4$. On negative outer gaps the positive part is bounded by $(R-\rho)^2/4$ after extracting $|t|$. Integration gives

$$3(\rho J_1 - J_2)_o \le \frac{3}{8}\{(\rho - B)_+^2 T_+ + (R-\rho)^2 T_-\}.$$

The available weighted outer budget is $D - mQ$, not the entire $D$. The cover intersects this budget with the individual bounds $T_+ \le b_{+,u} - m_l B_l^2$ and $T_- \le b_{-,u} - m_l A_l^2$.

For the signed phase term $\Omega = J_1 + (\mu_4 - 3)/6$, retain

$$-RT_-/2 \le J_{1,o} \le RT_+/2.$$

The central minimum of $P_1$ occurs at $B_l$ and an $A$ endpoint; its maximum occurs at $B_u$ and $A$ clipped to the interval at $A = B_u$. A nonnegative polynomial bound is multiplied by $m_u$ for an upper bound and $m_l$ for a lower bound; a negative one uses the opposite endpoint. For example, the binary endpoints $A = 3/2$, $B = 2/3$ make this maximum negative; the sign switch is necessary even when $D < 2$. The fourth-moment sign intervals also sharpen $K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2$. The complete $K_1, \Omega, \varphi$ rectangles are used in the signed expression $|cK_1 + \varphi\Omega|$, without replacing it by two separate absolute values.

**A compulsory first moment before each signed evaluation.**

The physical inequality $D \ge \sigma\rho - \gamma - 6m$ is especially useful inside a cell. If its prior ceiling is $D \le d$, and $\rho \ge l$, $\gamma \le \gamma_+$, then

$$m \ge m_* := \max\{0, (\sigma_l l - \gamma_+ - d)/6\}. \tag{66}$$

If $m_* > 1/2$, the cell is empty. Otherwise the common sign-moment inequality $4m^2 + \beta^2 \le 1$ forces $|\beta| \le \sqrt{1 - 4m_*^2}$. This restriction must be intersected with the beta interval for this gamma cell, not carried from a different gamma cell. Recompute all dependent sign-moment ranges after the intersection. These are necessary conditions on one law; no sufficiency or moment-realizability assertion is made.

For completeness, the event distance and conditional moment refinements fit the Fourier input exactly. On the acquired actual gap, use $\mathcal{E} = \{XX' < 0\}$, a downward square root of (65), and the larger of the two valid probability lower bounds

$$\alpha = \max\left\{\frac{(1 + \min|\beta|^2)^2}{2u^2}, \quad 2\min_{p=p_-,p_+} p(1-p)\right\}.$$

The conditional second-moment ceiling is the smaller of $2/\alpha$ and the corner maximum of the exact expression displayed above. Its numerators are positive, it increases with m, is affine in beta, and is convex in p, so these corner choices are valid. Each event option is inserted into all six HIGH terms. Taking the minimum with the complete universal-event bound preserves the same actual Fourier integral. None of these statements follows merely from declaring a moment rectangle feasible.

## D.7 The signed enclosure and its ordered update

On one closed moment cell retain bounds for $(\rho, \beta, \gamma, m)$, an already proved support radius R, an incoming raw ceiling T, and the weighted defect invariant $nD \le Nd$. Thus $D \le d$ for every actual $n \ge N$. Intersect all necessary moment conditions just proved, with directed endpoint operations. A positive numerator with a zero denominator proves incompatibility only through its original positive-semidefinite inequality; a zero-over-zero correction is dropped. No division is made before this check. Bisection of the monotone first-moment necessary inequality keeps its upper endpoint, so its rounding cannot remove a law.

Here is an explicit description of the coefficient enclosures. Let $M$ be the resulting upper fourth moment and let $[J_-, J_+]$ contain J; one can use

$$J_- = \max\{0, ((3 + \sigma_l)l - \gamma_+ - d)/3\}, \quad J_+ = \min\{u + 2m_+, ((3 + \sigma_u)u - \gamma_-)/3\},$$

with a rational $\sigma_u > \sqrt{10}$. For the beta term take the full corner maximum of $3\beta(J+\gamma)/2$. For the gap charge take the minimum of the complete coarse charge (60) and the central-plus-outer charge just proved. Write this bound as $G_*$ and the beta-term maximum as $B_*$. Let $P_* \geq 0$ be any lower bound for the maximum of the three independent fifth-moment penalties. Then a valid complete constant bound is

$$A = \min\{B_* + G_* + u^3/4 - l/2 - P_*, \quad B_* + 2uM - l/2\}. \tag{67}$$

The fourth semimoment inequalities and the signed product rectangle for $\rho\gamma$ give an interval $[k_-, k_+]$ containing $K_1 = \mathbb{E}X_+^4 - \mathbb{E}X_-^4 - 3\beta - 3\rho\gamma/2$. The central polynomial $mP_1$ and the signed outer bounds give an interval $[\omega_-, \omega_+]$ containing $\Omega = J_1 + (M_4 - 3)/6$. Intersect them with every previous valid interval. When multiplying by a moment endpoint, the sign of the coefficient is retained. These constructions explain every coefficient in the next quadratic, without assuming that unrelated corner maxima are attained by one common law.

Require the incoming raw gate $0 \leq T < Nc_l$. By Corollary 4.2,

$$\varphi = \phi(H/\sqrt{n}) \in [\phi_-, 2/5], \qquad \phi_- = \phi_l(1 - T/(Nc_l)) > 0.$$

For a formal coefficient $v \in [c_-, c_+]$ set

$$\begin{aligned} B_v &= K_F + vA, \\ F_v &= \max_{k\in\{k_-,k_+\},\ \omega\in\{\omega_-,\omega_+\},\ \psi\in\{\phi_-,2/5\}} |vk + \psi\omega| + s_F, \\ Q_v &= vl + \tfrac{1}{2}\min\{\phi_-\gamma_-, (2/5)\gamma_-\}. \end{aligned} \tag{68}$$

The actual raw remainder is at most $B_c + F_c|H| - Q_cH^2$. The maximum over the signed rectangle is valid by successive affine maximization, and the absolute value preserves convexity. For fixed $|H|$, this expression is convex in c: the constant and curvature parts are affine and $F_c$ is convex. Thus the two endpoints c-,c+ suffice, even if A is negative. A negative lower gamma must be priced with the upper density endpoint; it cannot be reflected inside the contact law.

**Lemma D.4** (One ordered signed step)**.** *Suppose all preceding cell bounds hold and $Q_v > 0$ at both endpoints. Let upward rounding be applied to every proposed numerical ceiling. Set*

$$\begin{aligned} T' &= \min\{T, \max_{v=c_-,c_+}(B_v + F_v^2/(4Q_v))\}, \\ \eta &= (c_l - T'/N)/2, \\ U' &= \max_{v=c_-,c_+}(B_v + F_v^2/[4(Q_v+\eta)]). \end{aligned}$$

*A negative T' or U' excludes the cell. Otherwise $\mathcal{R}_c \leq T'$, $\mathcal{U} \leq U'$, and the new weighted defect bound is*

$$nD \leq Nd', \qquad d' = \min\{d, 6U'/(N\phi_l)\}.$$

*A valid raw-threshold ceiling is*

$$|H| \leq \max_{v=c_-,c_+} \frac{F_v + \sqrt{F_v^2 + 4(Q_v+\eta)\max(0, B_v)}}{2(Q_v+\eta)}. \tag{69}$$

*Proof.* Complete the two raw squares first. The incoming strict gate persists under $T' \leq T$, and gives $\eta > 0$. Only now apply Corollary 4.2 and complete the two resource squares with curvature $Q_v + \eta$. A negative complete ceiling is incompatible with the nonnegative raw or resource expression. The resource contains $n\phi(0)D/6$, so the displayed weighted bound follows without setting the actual sample size equal to N. If the maximum of both resource quadratics at a phase were negative, no witness could have that phase. Replacing B by its positive part enlarges each positive root and gives (69). Previous valid resource or phase bounds may also be intersected, but an unlicensed resource bound cannot be fed back as a raw ceiling. □

### D.8 Closed coverage and the selected acquisition theorem

For each of the original 24 rho cells, acquire a support radius with the independent heat/ray test before recomputing higher moments. Each signed pass then partitions a full necessary beta enclosure into eight closed intervals, each resulting necessary gamma enclosure into eight closed intervals, and each necessary common-m enclosure into two closed intervals. The original beta interval is restored for every gamma slice before applying (66). Discarding an outer interval counts all its descendants, and requires one of the proved necessary conditions to be inconsistent. In every processed rho cell the active and excluded counts sum to 128.

Each surviving actual law satisfies the cell update of Lemma D.4. Pool upper bounds by maxima and lower bounds by minima before the next pass. This may enlarge the possible set, but cannot lose an actual witness. A rho cell disappears from later passes only when all of its descendants have been excluded. Degenerate closed intervals may duplicate boundary work; they do not leave holes.

For export, intersect each surviving necessary probability interval with all 100 closed panels $[j/100, (j+1)/100]$, $j = 0, \dots, 99$. A cell meeting a shared endpoint contributes to both panels. From $\theta = 1 - m^2/[p(1-p)]$ obtain a panel upper bound using its lower m endpoint and the maximum of $p(1-p)$ on the intersection, including $1/4$ if it crosses $1/2$. Pool this simultaneously with the raw, weighted defect, rho, beta, and phase bounds of the same cells.

**Certificate condition D.5** (The four signed passes)**.** Starting with Condition D.3, all support, moment, actual-event, Fourier, denominator, signed-curvature and ordered raw-license gates hold on the complete four-pass cover just described. Its exact rational evaluation contains 11264 nominal signed boxes in total. The complete 100-panel export satisfies, on every nonempty panel,

$$\begin{aligned} &\mathcal{R}_c \le 2.1995174, && nD \le 32.20665378, && D \le 0.76682509, \\ &\rho \le 71527/48000, && |\beta| \le 0.50549963, && |H| \le 1.83605680. \end{aligned}$$

Fourteen rho cells survive, and thirty probability panels are nonempty. The largest support radius among the surviving final rows is $157/50$; thus the export also supplies

$$\operatorname{supp} P \subset (-157/50, 157/50). \tag{70}$$

This is the named fixed-target coarse proof object, with its own targets and complete incidence requirements. Its reproduction is specified in Appendix E; numerical acceptance remains distinct from the analytic proof that its leaves enclose all possible witnesses. The preceding statements supply that reduction and explicitly identify the finite predicates used. They do not change an older certificate's hypotheses or its target.

*Proof of Theorem 5.1.* Theorem D.1 supplies the actual support before higher moments are used. Appendix A and the actual-event constructions prove the complete Fourier bounds on their specified domains. The coarse coefficient inequalities and the two ordered quadratic steps establish the first actual gap. Every signed cell thereafter uses the moment and physical-gap bounds of this same law, followed by Lemma D.4. Induction over the four passes preserves the weighted invariants. The complete closed cover and outward pooling prove that every witness reaches an incident output panel with the stated bounds. □

The argument contains no upper bound for the witness's actual score, no scalar score recurrence, and no second deleted-score assumption. It uses only the quadratic improvement in the threshold term. Sections 6 and 7 replace downstream panel arguments by one support region and two direct comparisons at $n \ge 700$. The internal entry cover remains intact. The independent finite base is a separate input.

# E Certificates and independent reproduction

The analytical hypotheses, the finite numerical verification, and the missing finite base are separate obligations. This appendix identifies the computation used by Theorem 5.1; it does not supply the base assumed in Theorem 8.2.

## E.1 What is stored and what is recomputed

The retained format and target are, respectively,

`FIXED_C_COARSE_ENTRY_V7`
`INITIAL42_FIXED_TARGET_COARSE_V7`.

The JSON object stores four lists of 24 moment-cell records and all 100 probability-panel records. A live cell records its support radius, remainder and defect ceilings, and active/excluded descendant counts; a previously excluded cell is marked accordingly. It does not store 11264 individual leaves. The checker reconstructs the subdivisions, exclusions, bounds, and panel incidences. The soundness argument is Proposition 5.2.

For example, the first signed pass records a checkpoint for cell index 0. The checker reconstructs its moment interval as $[1 + 3361i/96000, 1 + 3361(i+1)/96000]$ at index $i$; the JSON stores $i$, not these two endpoints. Its descendants and their enclosures must still be recomputed.

| Quantity for $\rho$-cell 0 | Value and provenance |
|---|---|
| Moment interval (reconstructed) | $[1, 99361/96000]$ |
| Support radius (stored) | $5/2$ |
| Remainder ceiling | $27071971/50000000 = 0.54143942$ |
| Defect ceiling $d$ | $18233099/100000000 = 0.18233099$ |
| Active / excluded descendants | 128 / 0 |

The last numerical bound means $nD \le 42d = 7.65790158$ for every actual $n \ge 42$ in the cell, rather than only for a law at index 42.

All paths below are relative to the repository root. The retained source and data can be read at the immutable baseline commit linked in Section 9; they are unchanged by this revision.

**Certificate**
`mathematics/fixed-target-v7/2026-10-27/entry-proof.json`

**Checker**
`mathematics/fixed-target-v7/2026-10-27/accept_entry.py`

**Source-isolated replay**
`mathematics/fixed-target-v8/2026-10-28/isolated_entry_replay.py`

**Selected exporter and exploratory producer**
`mathematics/fixed-target-v7/2026-10-27/export_entry.py`, `mathematics/fixed-target-v7/2026-10-27/signed_probe.py`

**Two-pass ancestor**
`mathematics/fixed-target-v6/2026-10-27/signed-two-pass-check.py` and its pinned dependencies.

The exploratory producer considers candidate subdivisions, support radii, and passes. The exporter selects the first four stages of a six-stage discovery trace and rounds ceilings outward. Acceptance reconstructs the two coarse and four signed passes; it does not rely on a discovery success label. The selected path uses integer and rational arithmetic. Generation and checking share mathematical kernels, so this is not a claim of independent implementations.

### E.2 Exact arithmetic and its limits

Rationals are represented by integer numerators and denominators. Outward rounding deliberately weakens a bound. Square-root endpoints are checked by rational square comparisons. For a nonnegative numerator, an upper numerator bound divided by a positive lower denominator bound gives an upper quotient bound. Signed quotients require the endpoint extrema: if $a \in [a_-, a_+]$ and $0 < b_- \le b \le b_+$, then

$$\frac{a}{b} \in \left[ \min_{i,j\in\{-,+\}} \frac{a_i}{b_j}, \max_{i,j\in\{-,+\}} \frac{a_i}{b_j} \right].$$

A denominator interval containing zero requires a separate argument; this rule does not permit division by it. The exponential helper uses, for rational $0 \le y \le \min(x, 64)$,

$$e^{-x} \le e^{-y} \le \left( \sum_{j=0}^{64} \frac{y^j}{j!} \right)^{-1},$$

followed by upward rounding. The positive exponential series proves this enclosure. The Gaussian constants, including the narrow interval for $C_{\mathrm{E}}$, are proved in Appendix B.

Numerical precision determines whether an enclosure is sharp enough to prove a strict inequality; extra digits alone do not prove it. Uniformity in $n$ comes from the analytical majorants in the Fourier appendix, not from sampling indices or assuming monotonicity of $C_n$. The checker semantics, their correspondence with the displayed formulas, and the integer/rational implementation remain part of the computational trust boundary. Hashes identify bytes, not mathematical truth. Subdivision certificates have established methodological precedents [17]; that precedent does not validate this custom format or transfer a proof assistant's guarantee to Python.

### E.3 Three separate checks

Use a fresh output path for each replay. From the repository root, the retained entry is reproduced by

```
python3 -I -B \
  mathematics/fixed-target-v8/2026-10-28/isolated_entry_replay.py \
  --output-dir runs/compact-v16-entry-review
```

The wrapper copies a fixed 13-file source/data set into an isolated source environment. The wrapper status is `PASS_ISOLATED_REPRODUCTION_FIXED_ENTRY_V7`; its numerical output is `PASS_INITIAL42_FIXED_TARGET_COARSE_V7`. It reconstructs 11264 nominal signed boxes, 100 panels, 30 nonempty panels and 14 surviving moment cells. Nineteen checker rejection controls and five wrapper controls test malformed or altered inputs. They supplement rather than replace the coverage proof.

The bounded historical checks and the current document's reference structure are reproduced separately by

```
python3 -I -B \
  manuscript/compact-v16/tools/recheck.py \
  compact-v16-bounded-review
```

Here the output path is relative to `manuscript/compact-v16/`; an absolute fresh path with an existing parent can instead be supplied. This runs nine unchanged v16 mathematical programs. They retain their original parameters, including the former small-noise minorant. They are regression evidence for the retained inputs, not verification of the revised scalar choices and not a replay of the coarse entry.

The changed scalar claims have their own exact checks:

```
python3 -I -B \
  manuscript/compact-v16/tools/check_revision_constants.py \
  --output-dir runs/compact-v16-scalar-review
```

This runs the geometry, count, and foundation checks. They reconstruct the rational comparisons from their definitions: the endpoint probability exclusion, six ray bounds, the new minorant's full rectangle argument, the common error envelope and its two quartic endpoints, the Fourier quadratic identity, and the narrow $C_{\mathrm{E}}$ enclosure. The continuous proofs of sign, concavity, monotonicity, and coverage remain in the text. Selected atomic-law examples are diagnostics, not an all-law proof. The revision record reports these executions separately from the inherited entry replay and from document compilation.

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